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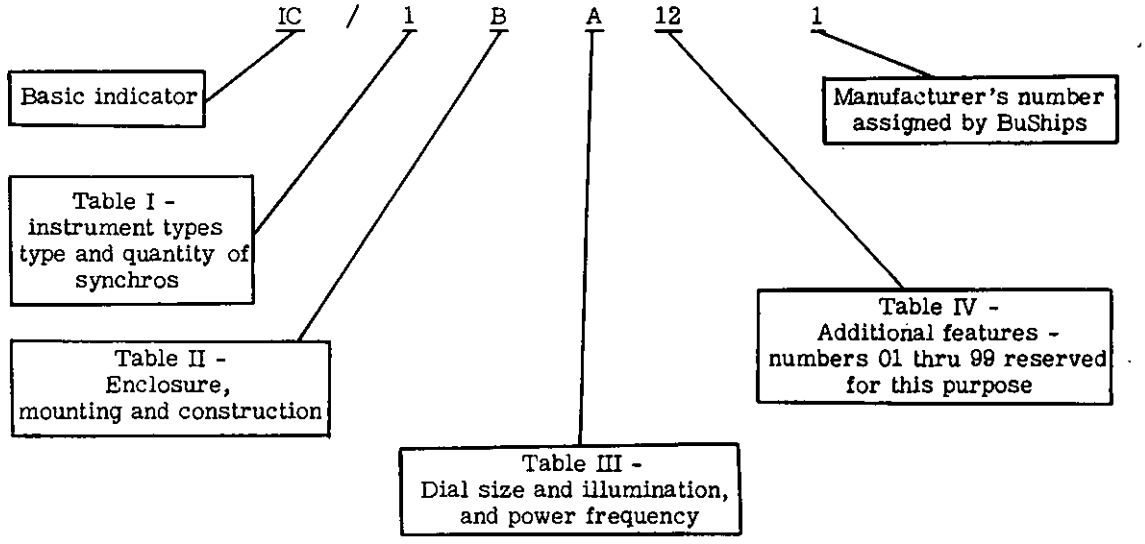
INTERIM MILITARY SPECIFICATION

SHIP CONTROL ORDER AND INDICATING EQUIPMENT - SELF-SYNCHRONOUS, (NAVAL SHIPBOARD USE)

1. SCOPE

1.1 Scope. - This specification covers self-synchronous order and indicating equipment for transmitting and receiving orders and indications for ship control purposes in Naval ships.

1.2 Classification. - The types of indicators, transmitters and indicator-transmitters shall be designated in the following form, as applicable (see 6.1):



1.2.1 Basic indicator. - The basic indicator for ship control order and indicating equipment is IC/.

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1.2.2 Type of instrument and type and quantity of synchro units. - The types and quantities of synchro receivers and synchro transmitters contained in the component shall be designated by a number symbol as shown in table I.

Table I - Type of instrument - type and quantity of synchro units.

Symbol	Type of instrument	Type and quantity of synchro units - Possible combinations
1	Single indicator	One 1F, or one 23TR6, or one 23TR4.
2	Double indicator	One 1F and one 3F, or two 23TR6, or one 23TR4 and one 31TR4, or one 2R, or one 1F and one 5F, or one 23TR4 and one 37TR4.
3	Triple indicator	Three 1F, or three 23TR6, or three 23TR4, or one 1F and two 3F, or one 23TR4 and two 31TR4, or one 2R and one 3F.
4	Single transmitter	One 5G, or one 37TX4, or one 6G.
5	Double transmitter	Two 5G, or two 37TX4, or two 3G, or two 31TX4.
6	Triple transmitter	Three 5G, or three 37TX4, or three 3G, or three 31TX4.
7	Single indicator-transmitter	One 1F and one 5G, or one 23TR6 and one 5G, or one 23TR6 and one 6G, or one 1F and one 1G, or one 2R, or one 23TR4 and one 37TX4, or one 23TR4 and one 31TX4.
8	Double indicator-transmitter	Two 1F and two 5G, or two 23TR6 and two 5G, or two 23TR6 and two 6G, or two 1F and two 1G, or two 2R, or two 23TR4 and two 37TX4, or two 23TR4 and two 31TX4.
9	Triple indicator-transmitter	Three 1F and three 5G, or three 23TR6 and three 5G, or three 23TR4 and three 37TX4.
10	Double indicator combined with single transmitter	One 5G and one 1F and one 3F, or one 5G and two 23TR6, or one 37TX4 and one 23TR4 and one 31TR4, or one 5G and one 2R, or one 5G and one 1F and one 5F, or one 37TX4 and one 23TR4 and one 37TR4, or one 5G and two 5F, or one 5G and two 3F, or one 37TX4 and two 31TR4.

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1.2.3 Mounting and construction. - The type of mounting and construction (see 6.1) of the enclosure shall be designated by a single letter as shown in table II.

Table II - Enclosure, mounting and construction.

Symbol	Mounting	Construction
B	Bulkhead	Cast
P	Panel	Cast
S	Pedestal	Cast
U	Universal	Cast
D	Deck	Sheet metal
W	Bulkhead	Sheet metal
X	Panel	Sheet metal
Y	Pedestal	Sheet metal
Z	Universal	Sheet metal

Note. - Cast enclosures are normally aluminum, except for rudder angle transmitters, bow plane angle transmitters, and stem plane angle transmitters which are bronze. Sheet metal enclosures are normally made of sheet brass, except where material scarcity compels the use of sheet steel.

1.2.4 Dial size, dial illumination and power frequency. - The characteristics of dial size, dial illumination and power frequency of the component shall be designated by a single letter as shown in table III.

Table III - Dial size, dial illumination and power frequency.

Symbol	Dial size	Dial illumination	Power frequency
			Cycles
A	4 inch diameter	yes	60
B	6 inch diameter	yes	60
C	Special	yes	60
D	4 inch diameter	no	60
E	6 inch diameter	no	60
F	Special	no	60
G	No dial	no	60
H	4 inch diameter	yes	400
I	6 inch diameter	yes	400
J	Special	yes	400
K	4 inch diameter	no	400
L	6 inch diameter	no	400
M	Special	no	400
N	No dial	no	400

Note. - Special dials include the following typical examples:

(a) 8 inch diameter dials for pedestal mounted engine order indicator-transmitters.
(b) 5 inch diameter dials for special compact engine order indicator-transmitters, and windows for observation of a selected section of propeller order dials.

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1.2.5 Additional features.- Special or additional features shall be designated by a number ranging from 01 through 99, reserved for this purpose as shown in table IV. Table IV is applicable to all indicators, transmitters, and indicator-transmitters.

Table IV - Instrument symbols and features.

Symbol	Type of instrument	Terminals for audible signals	Audible signals	Signal push-buttons	Pointer comb. E or F (see 3.3.6.2)	Pressure-proof	Dial range for propeller order single indicator	6G synchro	No additional features	Transmitter operation	Detents or friction brakes	Pilot light	Wrong direction contacts
01	Indicators	X											
02	Indicators	X	X										
03	Indicators	X		X									
04	Indicators	X	X	X									
05	Indicators	X			X								
06	Indicators				X								
07	Indicators	X		X			0 - 400 RPM						
08	Indicators	X	X	X			0 - 400 RPM						
09	Indicators	X		X			0 - 800 RPM						
10	Indicators	X	X	X			0 - 800 RPM						
11	Indicators								X				
12	Indicators					X							
20	Transmitters									Lever			
21	Transmitters									Shaft only			
22	Transmitters			X						Knobs	Friction brake	X	
23	Transmitters	X		X						Knobs	Detent & Friction		
24	Transmitters	X	X	X						Knobs	Detent & Friction		
25	Transmitters	X		X						Knobs	Detents		
26	Transmitters	X	X	X						Knobs	Detents		
27	Transmitters	X		X			0 - 400 RPM			Knobs	Friction brake		
28	Transmitters	X	X	X			0 - 400 RPM			Knobs	Friction brake		
29	Transmitters	X		X			0 - 800 RPM			Knobs	Friction brake		
30	Transmitters	X	X	X			0 - 800 RPM			Knobs	Friction brake		
31	Transmitters							X		Lever			
32	Transmitters							X		Shaft only			
40	Indicator-transmitter	X		X						Knobs	Detents	X	
41	Indicator-transmitter	X	X	X						Knobs	Detents	X	
42	Indicator-transmitter	X		X	X					Knobs	Detents	X	
43	Indicator-transmitter	X	X	X	X					Knobs	Detents	X	
44	Indicator-transmitter	X		X	X					Knobs	Detents	X	X
45	Indicator-transmitter	X	X	X	X					Knobs	Detents	X	X
46	Indicator-transmitter	X		X						Knobs	Detents		
47	Indicator-transmitter	X	X	X						Knobs	Detents		
48	Indicator-transmitter			X						Knobs	Friction brake	X	
51	Indicator-transmitter	X	X	X			0 - 400 RPM			Knobs	Friction & Detents	X	
52	Indicator-transmitter	X	X	X			0 - 800 RPM			Knobs	Friction & Detents	X	
53	Indicator-transmitter	X	X	X						Handle	Detents	X	
54	Indicator-transmitter	X		X				X		Knobs	Detents	X	
55	Indicator-transmitter	X	X	X				X		Knobs	Detents	X	
56	Indicator-transmitter	X		X	X			X		Knobs	Detents	X	
57	Indicator-transmitter	X	X	X	X			X		Knobs	Detents	X	
58	Indicator-transmitter	X	X	X				X		Handle	Detents	X	

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1.2.6 Manufacturer.- The number indicating the special, or additional, features shall be followed by a dash and a number indicating the manufacturer. This number will be assigned by the bureau or agency concerned at the time of action on drawings of the component.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 - Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).
NN-B-601 - Boxes, Wood-Cleated-Plywood, for Domestic Shipment.
NN-B-621 - Boxes, Wood, Nailed and Lock-Corner.
QQ-S-692 - Steel, Sheet, Cold Rolled.
QQ-S-781 - Strapping, Flat, Steel.
LLL-B-631 - Boxes, Fiber Corrugated (for Domestic Shipment).
LLL-B-636 - Boxes, Fiber, Solid (for Domestic Shipment).

MILITARY

JAN-P-104 - Packaging and Packing for Overseas Shipment - Crates, Sheathed, Wood, Nailed.
JAN-P-105 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Cleated, Plywood.
JAN-P-106 - Packaging and Packing for Overseas Shipment - Boxes, Wood, Nailed.
JAN-P-108 - Packaging and Packing for Overseas Shipment - Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
MIL-P-116 - Preservation, Methods of.
JAN-P-125 - Packaging and Packing for Overseas Shipment - Barrier-Materials, Waterproof, Flexible.
JAN-P-132 - Packaging and Packing for Overseas Shipment - Crates; Unsheathed, Wood; Nailed (for Maximum Net Load of 2,500 Pounds).
JAN-P-139 - Packaging and Packing for Overseas Shipment - Plywood, Container-Grade.
MIL-A-140 - Adhesive, Water-Resistant, Waterproof Barrier-Material.
MIL-B-233 - Boxes, Repair Parts.
JAN-P-658 - Packaging and Packing of Electrical Equipment and Spare Parts (Electronic, Electrical, and Electro-Mechanical).
MIL-C-915 - Cables, Electric (Shipboard Use).
MIL-I-983 - Interior Communication Equipment; General Specification for.
MIL-B-994 - Brass, Naval, Wrought.
MIL-C-2194 - Cables, Electric (Reduced Diameter Types) (Shipboard Use).
MIL-C-6057 - Crates and Boxes; Aircraft and Airframe Component Parts (General Specification for).
MIL-B-15071 - Books, Instruction; Preparation, Contents and Approval.
MIL-A-15303 - Audible Signals, Interior Communication and Small Craft (Bells, Buzzers, Horns and Sirens) (Naval Shipboard Use).
MIL-B-16541 - Bronze, Valve: Castings.
MIL-A-17129 - Aluminum Alloy Casting; Sand, Resistant to Salt Water.

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NAVY DEPARTMENT
General Specifications for Inspection of Material, and
Appendix VII - Welding.
Part C.
46B24 - Bronze: Castings, Ornamental.

STANDARDS

MILITARY
MIL-STD-24 - Revision of Drawings.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking of Shipments.

DRAWINGS

BUREAU OF SHIPS
9-S-5235-L - Terminal Tubes, Steel, Type 51, Machined and Drawn.
9000-S6202-73754 - Push Switch, W.T., Single Type E16A.
9000-S6202-73902 - Stuffing Tubes, P P Brass for Electric Cables.
9000-S6202-73903 - Molds and Arrangements for Sealing and Packing Cable Ends for P P Installations.
9000-S6504-73687 - Standard Dial Markings for I.C. Telegraph and Indicating Systems.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. - The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification:

CONSOLIDATED FREIGHT CLASSIFICATION COMMITTEE
Consolidated Freight Classification - Ratings, Rules and Regulations.

(Application for copies should be addressed to the Consolidated Freight Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Qualification. - The ship control order and indicating equipment furnished under this specification shall be a product which has been tested and has passed the qualification tests specified in section 4 (see 6.2).

3.2 General features. -

3.2.1 This ship control order and indicating equipment shall be in accordance with the following paragraphs of Specification MIL-I-983 in addition to the requirements specified hereinafter. Whenever a requirement of Specification MIL-I-983 conflicts with a requirement of this specification, the requirement of this specification shall govern.

Materials, general
Stressed or vital parts
Unstressed or nonvital parts
Wood
Magnesium
Aluminum
Brass and bronze
Iron and steel
Protection against corrosion (salt spray test is required)
Gaskets
Component dial windows
Plastics
Inflammable materials
Toxic compounds
Lubricants and lubrication

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Electrical parts, general
Resistors
Potentiometers
Transformers
Synchros
Fungus-inert materials
Restriction of materials
Electrical tapes (see 3.2.3.2)
Enclosures - general
Accessibility
Degree of enclosure (see 3.3.1)
Blastproofness
Enclosure mounting
Stiffening grooves
Minimum sheet thickness
Through bolting
Terminal tubes
Condensation drain
Cover bolts
Ventilation
Size (for submarines)
Temperature and humidity (special cold limit applies)
Threaded parts
Locking devices
Washers
Interchangeability and standardization
Shock, vibration and inclination (vital vibration test required)
Component dials and pointers
Painting
Designation and marking
Accelerated life
Design for production
Rounded corners and edges
Internal unit protection
Ball bearings
Drilled and tapped holes
Welding (see 3.2.1.1)
Primary power circuits
Power supply tolerances (see 3.4.1)
Personnel protection
Clearances (insulation resistance and dielectric strength)
Shielding and radio frequency noise interference
Audible noise
Grounds
Soldering
Mounting of parts
Internal assembly connection
Terminal strips, connectors and terminals
Wiring
Dial illumination
Lamps
Batteries
Accuracy
Drawings
Instruction books
Maintenance parts kit
Repair parts
Workmanship and general inspection
General information

3.2.1.1 Welding. - Welding shall be in accordance with Appendix VII, Part C of the General Specifications for Inspection of Material.

3.2.2 The ship control order and indicating equipment shall consist of indicators, transmitters, and indicator-transmitters to perform the following functions:

3.2.2.1 Indicators. - Each indicator shall contain, in a single enclosure, the mechanism necessary to receive and display the synchro order or information data transmitted by the associated transmitter. Indicators shall be designated as single, double or triple (see 3.5).

3.2.2.2 *Transmitters.* - *Each transmitter shall contain, in a single enclosure, the mechanism necessary* to receive a mechanical input (order or information), and to transmit a synchro signal corresponding to the mechanical input; or to receive a mechanical input (order or information), transmit a synchro signal corresponding to the mechanical input, and display the order or information transmitted. Transmitters shall be designated as single or double (see 3.6).

3.2.2.3 Indicator-transmitters. - Each indicator-transmitter shall contain, in a single enclosure, the characteristics specified in 3.2.2.1 and 3.2.2.2 for indicators and transmitters, respectively. Indicator-transmitters shall be designated as single, double or triple (see 3.7).

3.2.3 Special material features. -

3.2.3.1 Electrical tapes. - The use of electrical tapes shall be held to a minimum.

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3.2.3.2 Push buttons. - Push buttons shall incorporate internal units in accordance with Drawing 9000-S6202-73754.

3.2.3.3 Audible signals. - Bells and buzzers shall conform to Specification MIL-A-15303. The use of bells shall be limited to types B1S4, B2S4, B3S4, B1F4, B2F4, and B3F4 as specified herein. The use of buzzers shall be limited to types Z1S4 and Z1F4 as specified herein.

3.2.3.4 Enclosure materials. - Material for enclosure cases and covers, and associated pedestals where specified, shall be in accordance with one of the following materials, as designated by the instrument type number (see 1.2.3).

3.2.3.4.1 Cast aluminum (class 3 or 4). - Cast aluminum (class 3 or 4) shall be in accordance with Specification MIL-A-17129.

3.2.3.4.2 Sheet brass. - Sheet brass shall be in accordance with Specification MIL-B-994.

3.2.3.4.3 Sheet steel. - Sheet steel shall be in accordance with Specification QQ-S-692.

3.2.3.4.4 Cast bronze. - Cast bronze shall be in accordance with Specification MIL-B-16541 for pressureproof enclosures.

3.2.4 Drawings. - In addition to the Military-Standard specified in Specification MIL-I-983, Standard MIL-STD-24 shall also be applicable.

3.3 Mechanical features. -

3.3.1 Degree of enclosure. - Except for the pressureproof rudder angle indicator specified in 3.5.1.6.4, all enclosures shall be watertight (3 feet).

3.3.2 Compactness of design. - The equipment shall be as compact as practicable, consistent with sound engineering principles of design necessary to meet the requirements of this specification. The limiting dimensions specified herein shall not be exceeded.

3.3.3 Location of operating attachments. -

3.3.3.1 Knob operated instruments. - All operating knobs, dimmer knobs, signal push buttons, and pilot lights shall be located on the enclosure surface which contains the dial window(s).

3.3.3.1.1 Condensation drain. - The condensation drain for panel mounted and universal mounted instruments shall be located at the lower edge of the enclosure surface referred to in 3.3.3.1. The condensation drain for bulkhead mounted instruments shall be located in the bottom surface (not mounting surface) of the instrument. The condensation drain for pedestal mounted instruments shall be located in the terminal tube plate, and the outlet of the drain shall be located in the outer (exposed) surface of the terminal tube plate.

3.3.3.2 Handle operated instruments. - All dimmer knobs, signal push buttons, and pilot lights shall be located on the aft side of the instrument housing.

3.3.3.2.1 Condensation drain. - The condensation drain shall be located in the terminal tube plate, and the outlet of the drain shall be located in the outer (exposed) surface of the terminal tube plate.

3.3.4 Operating knobs. - The stem of each operating knob shall be provided with a smooth skirt cast integral with the knob, and which shall completely cover the gland nut of the shaft stuffing tube, or shall be provided with other means to prevent cutting the operator's knuckles. The operating knobs shall clear the surface of the enclosure by at least 0.80 inch, and shall be shaped to provide a firm and positive grip for the operator. The operating knobs shall be of cast bronze in accordance with Specification 46B24.

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3.3.4.1 Removable knobs. - Transmitter operating knobs, if removable with the cover, shall have positive means for engaging the drive mechanism when secured in the operating position.

3.3.5 Watertight glands. - Watertight glands shall be provided for all operating shafts extending through the enclosure.

3.3.6 Pointers. - Pointers shall be of sheet aluminum, sheet brass, or of plastic, dependent upon the method used for illumination of the pointers. When sheet aluminum or sheet brass is used the pointers shall have rolled or bevelled edges so as to contribute to the reflection of light provided for the pointer illumination by the basic equipment illumination.

3.3.6.1 Shape. - Where two or more concentric pointers are incorporated in the equipment, the pointers shall be designed so as to provide ready and positive identification by the differences in their shapes.

3.3.6.2 Color. - The color of the pointers and markings shall be one of the combinations listed in table V as specified herein.

Table V - Color of pointers and markings.

Combination	Number of pointers	Dial Illumination	Single pointer		Inner pointer		Middle pointer		Outer pointer	
			Pointer	Marking	Pointer	Marking	Pointer	Marking	Pointer	Marking
A	1	Yes	White	Black	---	---	---	---	---	---
B	1	No	Black	White	---	---	---	---	---	---
C	2	Yes	---	---	Green[1]	White	---	---	Red	Black
D	2	No	---	---	Green[1]	White	---	---	Red	White
E	2	Yes	---	---	Red [1]	Black	---	---	White	Black
F	2	No	---	---	Red [1]	White	---	---	Black	White
G	3	Yes	---	---	White[1]	Black	Green[1]	White	Red	Black
H	3	No	---	---	Black	White	Green[1]	White	Red	White
I	3	Yes	---	---	Green[1]	White	White[1]	Black	Red	Black
J	3	No	---	---	Green[1]	White	Black	White	Red	White

[1]A distinctive black and white border shall be provided for these pointers.

3.3.7 Pedestal mountings. - Pedestals shall be of the same material as that required for the equipments and will be determined by the type designation.

3.3.7.1 Access to terminals. - Terminal access, where practicable, shall be obtained by removal of the cover, or covers; and the terminals and ship's wiring shall be accessible without removing any internal parts or units.

3.3.7.1.1 Removable cover plate. - Where dimension limitations preclude the provision of terminal access in accordance with 3.3.7.1, a removable cover plate shall be incorporated in all pedestal mounted equipments to provide adequate access to terminal strips for connection of ship's wiring. The removable cover plate, where provided, shall be located above the terminal tube plate.

3.3.7.2 Condensation drain. - The condensation drain for pedestal mounted equipments shall be located in the terminal tube plate.

3.3.7.3 Terminal tube plate. - The terminal tube plate shall not be drilled and tapped for ship's cable entrance. (Drilling and tapping of the terminal tube plate will be done by the installing activity.) The terminal tube plate shall be of sufficient thickness to provide adequate support for the ship's cable(s) when installed.

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3.3.7.4 Audible signals on pedestal mounted equipments. - Audible signals attached to pedestal mounted equipments shall be located so as not to interfere with their normal operation and not to increase the overall space requirements of the equipment. The mounting of the audible signal on the equipment shall be such that the wiring to the audible signal shall be taken directly through the enclosure and into the audible signal. The audible signal(s) shall be attached to the enclosure in a manner so as to incorporate the wiring to the audible signal and maintain the watertight integrity of the enclosure.

3.3.8 Bulkhead, panel, and universal mounted equipments with cast enclosures. - With the exception of the pressure-proof rudder angle indicator specified herein, equipments shall be provided with cable entrance holes drilled, tapped, and plugged in accordance with Specification MIL-I-983. Entrance holes shall be provided in the bottom of each cast enclosure, and in the top of some cast enclosures, as specified herein. Entrance holes shall be drilled and tapped to accommodate terminal tubes required for the smaller size cable as specified hereinafter for each specific equipment type. However, adequate boss areas shall be provided in the bottom and top surfaces of each enclosure to provide for terminal tubes required for the larger size cable specified. For terminal tube sizes required for different sizes of cable, see Drawing 9-S-5235-L.

3.3.9 Bulkhead, panel, and universal mounted equipments with sheet metal enclosure. - These equipments shall be provided with removable plates in the bottom surface of the enclosure in accordance with Specification MIL-I-983. Cable entrance holes will be drilled in these removable plates by the installing activity. No cable entrances will be made through the top surface of the enclosure. The removable plate shall be provided with adequate area to provide for terminal tubes required for the larger size cable specified for each equipment type. For terminal tube sizes required for different sizes of cable see Drawing 9-S-5235-L.

3.3.10 Internal assemblies. - With the exception of the single transmitters specified in 3.6.1.1, 3.6.1.2, and 3.6.1.4, all equipment shall be provided with removable internal assemblies. Each internal assembly shall be so mounted that no internal operating parts are disturbed when the unit is removed from the enclosure. Clamping screws shall be accessible for adjustment of the synchro transmitter(s) or receivers. Protective extensions and hand grips for removal of the internal assembly shall be of sufficient length, and so spaced, as to provide a level mounting when the internal assembly is placed on a work bench in either the upright or inverted position.

3.4 Electrical features. -

3.4.1 Power supply. - The normal power supply of the equipment shall be 115 volts, single phase, with the frequency as designated in the type designation. (Frequency will be specified in the contract or order.)

3.4.2 Synchros. - Each synchro transmitter and synchro receiver shall be adjustable so that the associated pointer, dial, or instrument operating shaft (for instruments without a dial) indicates the following position when the synchro unit is at electrical zero:

Fixed dial	Pointer indication
Anchor order	Center of "OFF"
Engine order	Center of "STOP"
Motor order	Center of "STOP"
Plane angle	0
Propeller order	0
Rudder angle	0
Rudder order	0
Steering order	Center of "CORS" ("COURSE")

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Rotating dial	Dial indication
Course order	0
Propeller order	0

Transmitter (no dial)	Position of basic motivation
Plane angle	Zero angle position of plane
Rudder angle	Zero angle position of rudder

3.4.2.1 Loading characteristics of synchro units. - The maximum loading of a type 5G or 37TX4 synchro transmitter shall be limited to: (a) twelve type 1F or 23TR4 synchro receivers, respectively; or (b) nine type 3F or 31TR4 synchro receivers, respectively; or (c) six type 5F or 37TR4 synchro receivers, respectively; or (d) appropriate combinations of type 1F, 3F, and 5F synchro receivers or type 23TR4, 31TR4, and 37TR4 synchro receivers, respectively.

3.4.3 Pilot lights. - Each pilot light shall be visible through a small watertight window in the instrument enclosure. The color of the light, as seen by the operator, shall be red in accordance with the red light requirements of Specification MIL-I-983. The pilot light shall have sufficient intensity of brightness to be seen under daylight operating conditions, but shall not be objectionably bright when seen under night operating conditions (darkened spaces). A transformer shall be used in conjunction with one of the following Navy type lamps:

TB11
TB14
TS130
TS131
TS132

The lamp shall be energized from the secondary winding of the transformer. The primary winding of the transformer shall be connected across the power supply of the synchro transmitter designated for the applicable instrument.

3.4.4 Characteristics of instrument illumination. -

3.4.4.1 The instrument dials shall be illuminated by the "Duo-panel" system, a special form of edge illumination in which a translucent plastic outer panel (dial) is substantially isolated from intimate surface contact (optical contact) with a transparent plastic light-conducting panel.

3.4.4.1.1 The outer panel shall present a conventional daylight appearance, i.e., its front surface shall have "white" markings, numerals, etc., on an opaque dull black background. While the (diffuse) daylight reflectivity of the "white" markings shall be sufficiently high to result in adequate contrast with the background and good resulting legibility, the (diffuse) transmission of these markings shall be sufficiently high to permit the successful use of the outer panel in the "Duo-panel" system.

3.4.4.1.2 The outer panel shall not exceed 1/16 inch in thickness, and shall be: (a) of a thermo-setting plastic material such as a translucent, graphic, melamine-formaldehyde impregnated, paper-base laminate, or (b) of a high softening temperature acrylic material, having engraved (depressed) or embossed (relief) markings, and having an opaque black background, applied over a translucent white undercoating. The rear surface of this panel may be embossed with a screen or pebbled pattern to permit physical supporting contact while reducing to unimportance the areas of optical contact between it and the associated light conducting panel, if the use of spacers is not desirable.

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3.4.4.1.3 The light conducting panel shall be of 1/4 inch thickness, high softening temperature, colorless, transparent, acrylic sheet, to the rear surface of which small areas of compatible white lacquer, or enamel, have been applied, as necessary, to permit light to escape from the panel and illuminate the markings on the outer panel. This paint shall be selectively applied by any convenient silk screening or stenciling means capable of controlling the areas of the individual paint dabs in accordance with a predetermined, experimentally derived, pattern known to result in uniform illumination of all dial markings.

3.4.4.1.4 Light shall be obtained from lamps (see 3.4.4.1.5.1.1) so located as to introduce light into the conducting panel, and provided with individual cylindrical red plastic (high temperature) filters of spectral transmission characteristics as specified in Specification MIL-I-983. These filters shall be so designed that there is no escape of white light into any part of the illuminated dial or instrument.

3.4.4.1.5 Instrument pointers shall be either: (1) of metal in accordance with Specification MIL-I-983, or (2) of a thermo-setting plastic material or acrylic material, as specified in 3.4.4.1.2.

3.4.4.1.5.1 Metal pointers, when used, shall be illuminated by light distributed across the face of the dial at angles approaching the parallel to the plane of the dial. This light may be distributed by reflection from the inner surface of a cylindrical or rectangular metal accessory, from surfaces of the enclosure, or by an auxiliary plastic illuminator assembled above, but close to, the surface of the outer dial. Baffles, or louvers, and bezels shall be provided to reduce to a minimum all glare occasioned by the auxiliary pointer illumination, and to entirely prevent direct view of any lamp filament, reflected image of a lamp filament, luminous surface of a plastic illuminator, or other disturbingly bright object.

3.4.4.1.5.1.1 The means of illuminating the pointer preferably should utilize light from the same lamps that are used to illuminate the dial, but, if necessary, and particularly in larger instruments, separate lamps may be used as required.

3.4.4.1.5.2 Plastic or acrylic pointers, when used, shall be illuminated by light conducted through the pointer hubs from the light conducting inner panel, thus affording an indirect pointer illumination. The provisions of 3.4.4.1.5.1.1 shall apply.

3.4.4.1.6 The lamps shall be energized from the secondary of a transformer, and the lighting circuit shall be equipped with a rheostat, or other control device, to control light intensity from maximum value to full extinction when all lamps, or 50 percent of the lamps, are operative. At 115 volts input to the transformer, and with all lamps operative, the lamp socket voltages shall not exceed the following values calculated for 1000-hour life.

Navy type	Volts
TB11	7.5
TB14	6.6
TS53	6.87
TS52	6.87

3.4.4.1.6.1 The transformer shall have minimum voltage regulation consistent with reasonable compactness.

3.4.5 Accuracy. - All indicator pointers, indicator rotating dials etc. shall follow the associated transmitter setting within one geometrical degree for all variations of supply line voltage and frequency when tested in accordance with 4.3.2.

3.4.6 Oscillations. - When tested as specified in 4.3.3 the indicator pointer or indicator dial shall come to rest at the position synchronous with the transmitter within 3 seconds.

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3.5 Indicators.-

3.5.1 Indicators, single.- The pointer of each single indicator shall be positioned directly by a type 1F or type 23TR4 synchro receiver. Pointer combination A or B as specified in table V shall be required as applicable. When the type designation specifies that dial illumination shall be incorporated in the indicator, the illumination power supply shall be taken from the power supply of the synchro receiver. The terminal strip in each single indicator shall be provided with 6 terminals. (A type MSCA-7 cable in accordance with Specification MIL-C-2194 or a type MHFA-7 cable in accordance with Specification MIL-C-915 will be brought into the bottom of the single indicators for connection of ship's wiring, except for the propeller order indicator (see note following 3.5.1.5.5). A type DSGA-3 cable in accordance with Specification MIL-C-2194 or a type DHFA-3 cable in accordance with Specification MIL-C-915 will be brought into the top (see 3.3.8) or bottom (see 3.3.9) of bulkhead, panel, and universal mounted instruments for connection to the audible signal associated with the applicable instrument.)

3.5.1.1 Indicator, engine order.- The engine order indicator shall be provided with a 4-inch or a 6-inch dial, as designated by the type specified in the contract or order. No provision shall be made for pedestal mounting. (One cable for audible signal will be furnished (see 3.5.1).)

3.5.1.1.1 Size and weight.- The overall dimensions of the indicator, including the knob for control of the dial illumination, where designated, and the weight of this indicator shall not exceed the following:

	4 inch dial	6 inch dial
Height	8-1/2 inches	10 inches
Width	7 inches	9 inches
Depth	7 inches	7 inches
Weight	15 pounds	22 pounds

3.5.1.1.2 Dial marking.- Dial markings, lettering and graduations shall be in accordance with figure 1 of Drawing 9000-S6504-73687.

3.5.1.1.3 Terminal marking.- Terminals marked MB, MBB, MB1, MB2, and MB3 shall be provided for connection to the synchro receiver. The sixth terminal shall be provided with the double marking MB7, MB8. (An audible signal, when used in conjunction with the indicator, will be connected to the terminal marked MB7, MB8 and to either terminal MB or terminal MBB.)

Note.- The engine order indicators now active are as follows:

Active types

IC/1UA01
IC/1UD01
IC/1UE01

3.5.1.2 Indicator, motor order.- The motor order indicator shall be identical to the indicator specified in 3.5.1.1, except for dial markings.

3.5.1.2.1 Dial marking.- Dial markings, lettering, and graduations shall be in accordance with figure 4 of Drawing 9000-S6504-73687.

Note.- The motor order indicators now active are as follows:

Active types

IC/1UA01
IC/1UB01

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3.5.1.3 Indicator, plane angle, bow.- The plane angle, bow indicator shall be provided with a 4-inch dial, or a 6-inch dial, or a special dial, as designated by the type specified in the contract or order. No provision shall be made for a pedestal mounting.

3.5.1.3.1 Four inch and 6-inch dials.- The size and weight shall conform to 3.5.1.1.1.

3.5.1.3.1.1 Dial marking.- Dial markings, lettering, and graduations shall be in accordance with figure 8 of Drawing 9000-S6504-73687.

Note.- The plane angle bow indicators now active are as follows:

Active types
IC/1UA11
IC/1UB11

3.5.1.3.2 Special dial.- The overall dimensions of the indicator, including the knob for control of the dial illumination, where designated, and the weight of the indicator shall not exceed the following:

Height - 10 inches.
Width - 7 inches.
Depth - 8 inches.
Weight - 18 pounds.

3.5.1.3.2.1 Dial marking and dial size.- Only the active half of the dial shown on figure 8 of Drawing 9000-S6504-73687 shall be incorporated in the indicator. The radius of the active half dial shall be 4 inches.

3.5.1.3.2.2 Synchro receiver.- A type 3F or type 31TR4 synchro receiver shall be used for positioning the pointer. The synchro receiver shall be geared to the shaft of the pointer to facilitate compactness in the design of the indicator. The motion transmitted between the synchro receiver gear and the pointer gear shall be in a 1 to 1 ratio. The gear attached to the synchro receiver shall be provided with stops at points corresponding to apparent 35 degree indications of the indicator dial. Drawings and instruction books shall contain notes concerning the precautions to be taken at the time of initial installation, energization, and adjustment.

Note.- Plane angle bow indicators with special dials now active are as follows:

Active types
IC/1UC11
IC/1UF11

3.5.1.3.3 Pointer marking.- The pointer shall be engraved, or otherwise marked "BOW". The marking, "BOW", shall be read from left to right when the pointer indicates the zero position on the dial.

3.5.1.3.4 Terminal markings.- Terminal markings NB, NBB, NB1, NB2, and NB3 shall be provided.

3.5.1.4 Indicator, plane angle, stern.- The plane angle stern, indicator shall be identical to the indicator specified in 3.5.1.3, except for dial, pointer, and terminal markings.

3.5.1.4.1 Dial marking.- Dial markings, lettering, and graduations shall be in accordance with figure 9 of Drawing 9000-S6504-73687.

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3.5.1.4.2 Pointer marking. - The pointer shall be engraved, or otherwise marked, "STERN". The marking, "STERN", shall be read from left to right when the pointer indicates the zero position on the dial.

3.5.1.4.3 Terminal markings. - Terminal markings NS, NSS, NS1, NS2, and NS3 shall be provided.

Note. - Plane angle stern indicators now active are as follows:

Active types
IC/1UA11
IC/1UB11

3.5.1.5 Indicator, propeller order. - The propeller order indicator shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type.

3.5.1.5.1 Size and weight. - The size and weight of the indicator shall be as specified in 3.5.1.1.1 when bulkhead, panel or universal mounting is designated. When a pedestal mounting is designated, the size and weight of the basic indicator shall conform to 3.5.1.1.1 and the pedestal mounting shall conform to Specification MIL-I-983. The combined weight of the indicator and the associated pedestal shall not exceed the following:

4 inch dial - 80 pounds.
6 inch dial - 90 pounds.

3.5.1.5.2 Dial marking. - Dial markings, lettering, and graduations shall be in accordance with figure 20 (0-800 r.p.m.), or 22 (0-400 r.p.m.), Drawing 9000-S6504-73687, dependent upon the dial range designated by the type specified in the contract or order. The words "PROPELLER ORDER" shall be included on the dial, and located below the center of the dial. The backing signal, shown above "r.p.m." in the above figures shall not be incorporated in the dial of the indicator.

3.5.1.5.3 Audible signal. - A type B3S4 or B3F4 bell, shall be attached to the indicator when a pedestal mounting is designated by the type.

3.5.1.5.4 Terminals. - The terminal strip in the indicator shall be provided with 8 terminals.

3.5.1.5.5 Terminal markings. - Terminals marked M, MM, M1, M2, and M3 shall be provided for connection to the synchro receiver. Terminal markings M7 and M8 shall be provided for the sixth and seventh terminals, respectively. An audible signal, when used in conjunction with the indicator, shall be connected across terminals MM and M7. The signal pushbutton incorporated in the indicator shall be connected across terminals M and M8.

Note. - Cables for connection to ship's wiring and to audible signal - A type MSCA-10 cable (see Spec. MIL-C-2194) or a type MHFA-10 cable (see Spec. MIL-C-915) will be brought into the bottom of the instrument for connection of ship's wiring. A DSGA-3 cable (see Spec. MIL-C-2194) or a DHFA-3 cable (see Spec. MIL-C-915) will be used for connection to the audible signal associated with the instrument. Propeller order indicators now active are as follows:

Active types
IC/1UD07
IC/1UE07

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3.5.1.6 Indicator, rudder angle.- The rudder angle indicator shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type specified in the contract or order. The pressureproof indicator shall be provided with a 4-inch dial, dial illumination (but with dial illumination dimmer contained in a separate enclosure), and bulkhead mounting only.

3.5.1.6.1 Size and weight.- The provisions of 3.5.1.5.1 shall apply to all types of this indicator, except the pressureproof indicator. The overall dimensions of the pressureproof indicator and the weight shall not exceed the following:

Height - 10-3/4 inches
Width - 9 inches
Depth - 8 inches
Weight - 50 pounds

3.5.1.6.2 Dial marking.- Dial markings, lettering, and graduations shall be in accordance with figure 7 of Drawing 9000-S6504-73687.

3.5.1.6.3 Pointer marking.- The pointer shall be engraved, or otherwise marked, "ANGLE". The marking, "ANGLE", shall be read from top to bottom when the pointer indicates the zero position on the dial.

3.5.1.6.4 Special features of pressureproof indicator.- The openings in the case and cover shall be kept to a minimum. A pressureproof gland assembly shall be incorporated integral with the case, for passage of the cable entering the case. This gland assembly shall be located in the bottom face of the indicator enclosure, and shall be designed in accordance with Drawing 9000-S6202-73902 to accommodate molded rubber packing in accordance with Drawing 9000-S6202-73903 for the MHFF-10 cable (see 3.5.1.6.4.2). The dial window shall be made pressureproof by means of flat rubber gaskets forming seals between the outer and inner surfaces of the glass and the cover. The enclosure shall be of cast bronze in accordance with Specification MIL-B-16541. Walls of the enclosure shall not be less than 0.40 inch thick. The indicator shall withstand the depth charge test specified in Specification MIL-I-983.

3.5.1.6.4.1 Dial illumination dimmer for use with pressureproof indicator.- The dimmer shall be contained in a watertight (3-foot) enclosure designed for bulkhead mounting. (The dimmer will be installed remote to the pressureproof indicator, and inside the pressure hull.) The dimmer enclosure shall be of cast bronze in accordance with Specification 46B24. The overall dimensions of the dimmer enclosure, including the control knob, shall not exceed a 3-1/2-inch cube. The weight shall not exceed 4 pounds. Terminals marked N4 and N5 shall be provided for connection to the dimmer rheostat.

3.5.1.6.4.2 Cable for connection to ship's wiring.- An MHFF-10 cable (see Spec. MIL-C-915) will be brought into the bottom of the pressureproof instrument for connection of ship's wiring.

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3.5.1.6.5 Terminal markings.- Terminal markings N, NN, N1, N2, and N3 shall be provided in all indicators for connection to the synchro receiver. The pressureproof indicator shall be provided with additional terminals marked N4 and N5. Terminal N4 shall be connected to one terminal of the secondary winding of the illumination transformer. Terminal N5 shall be connected to one side of each of the dial illumination lamps. (The remaining terminal of the transformer secondary shall be connected to the opposite side of each dial illumination lamp.)

Note.- Rudder angle indicators now active are as follows:

Active types

Watertight indicators

IC/1UA11
IC/1UB11
IC/1UD11
IC/1UE11

Pressureproof indicator

IC/1BA12

3.5.1.7 Indicator, rudder order.- The rudder order indicator shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type. The size and weight and the dial marking shall be as specified in 3.5.1.5.1 and 3.5.1.6.2, respectively. (One cable for audible signal will be furnished (see 3.5.1).)

3.5.1.7.1 Pointer marking.- The pointer shall be engraved, or otherwise marked, "ORDER". The marking, "ORDER", shall be read from top to bottom when the pointer indicates the zero position on the dial.

3.5.1.7.2 Audible signal.- A type B2S4 or B2F4 bell, shall be attached to the indicator when a pedestal mounting is designated by the type.

3.5.1.7.3 Terminal markings.- Terminals marked L, LL, L1, L2, and L3 shall be provided for connection to the synchro receiver. Terminal marking L4 shall be provided for the sixth terminal. An audible signal, when used in conjunction with the indicator, shall be connected across terminals LL and L4.

Note.- Rudder order indicators now active are as follows:

Active types

IC/1UD01
IC/1UE01

3.5.2 Indicators, double.- With the exception of the course-steering order indicator specified in 3.5.2.1, and the propeller order indicator specified in 3.5.2.6, each double indicator shall be provided with two concentric pointers. The outer pointer shall be positioned directly by a type 1F or 23TR4 synchro receiver. Unless specified otherwise in the contract or order, the inner pointer shall be positioned by a type 3F or 31TR4 synchro receiver. The inner pointer shall be geared to the associated synchro receiver, and the motion transferred between the synchro receiver gear and the inner pointer gear shall be in a 1 to 1 ratio. When the type designation specifies that dial illumination shall be incorporated in the indicator, the illumination power supply shall be taken from the power supply of the type 1F or type 23TR4 synchro

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receiver. Unless otherwise specified herein, each double indicator shall be provided with two terminal strips of 6 terminals each. (A type MSCA-14 cable (see Spec. MIL-C-2194) or a type MHFA-14 cable (see Spec. MIL-C-915) will be brought into the bottom of the double indicators for connection of ship's wiring. DSGA-3 cable(s) (see Spec. MIL-C-2194) or DHFA-3 cable(s) (see Spec. MIL-C-915) will be brought into the top (see 3.3.8) or bottom (see 3.3.9) of bulkhead, panel, and universal mounted instruments for connection to the audible signal(s) associated with the applicable instrument.)

3.5.2.1 Indicator, course-steering order. - The course-steering order indicator shall be provided with a 4-inch or a 6-inch dial, as designated by the type. (One cable for audible signal will be furnished (see 3.5.2).)

3.5.2.1.1 Size and weight. - The overall dimensions of the indicator, including the knob for control of the dial illumination, where designated, and the weight of the indicator shall not exceed the following:

Bulkhead, panel, and universal mountings

	4-inch dial	6-inch dial
Height	10-1/4 inches	12 inches
Width	7 inches	9 inches
Depth	8-5/8 inches	8-5/8 inches
Weight	25 pounds	33 pounds

3.5.2.1.1.1 Pedestal mounting. - The overall dimensions of the indicator provided with a pedestal mounting shall be in accordance with 3.5.2.1.1 and Specification MIL-I-983. The combined weight of the indicator and the associated pedestal shall not exceed the following:

4-inch dial - 90 pounds
6-inch dial - 100 pounds

3.5.2.1.2 Dial marking. - Dial markings, lettering, and graduations shall be in accordance with figure 10 of Drawing 9000-S6504-73687. The inner portion of the dial, marked for steering orders and including the fixed reference point above "CORS", shall be fixed. The outer portion of the dial, marked for course orders, rotates about the fixed portion of the dial.

3.5.2.1.3 Pointer. - A single pointer shall be provided for indication of the steering order. The pointer shall be positioned directly by a type 1F or 23TR4 synchro receiver. Pointer combination A or B (see table V) shall be provided.

3.5.2.1.4 Course dial. - Indication of the course order shall be provided by means of rotation of the course dial ring, relative to the fixed indication mark shown above "CORS" on figure 10 of Drawing 9000-S6504-73687. The course dial ring shall be positioned by a type 5F or 37TR4 synchro receiver. The course dial ring shall be geared to the associated synchro receiver, and the motion transferred between the synchro receiver gear and the course dial gear shall be in a 1 to 1 ratio.

3.5.2.1.5 Audible signal. - A type B1S4 or B1F4 bell, shall be attached to the indicator when a pedestal mounting is designated by the type.

3.5.2.1.6 Terminal markings. - Terminals marked LA/C, LAA/C, LA1, LA2, and LA3 shall be provided for connection to the type 5F or 37TR4 synchro receiver associated with the course order indication. Terminals marked LA/S, LAA/S, LA4, LA5, and LA6 shall be provided for connection to the type 1F or type 23TR4 synchro receiver. Terminals marked LA7 and LA8 shall be provided for connection

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to the signal pushbutton and audible signal, respectively. The signal pushbutton incorporated in this indicator shall be connected across terminals LA and LA7. An audible signal, when used in conjunction with the indicator, shall be connected across terminals LAA and LA8.

Note. - The course-steering order indicators now active are as follows:

Active types
IC/2UB03
IC/2SB04

3.5.2.2 Indicator, engine order. - The engine order indicator shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type. No provision shall be made for a pedestal mounting. The indicator shall normally be provided with pointer combinations C or D (depending on whether or not internal dial illumination is specified) unless pointer combination E or F are designated (see table V). (Two cables for audible signals will be furnished (see 3.5.2).)

3.5.2.2.1 Size and weight. - The overall dimensions of the indicator, including the knob for control of the dial illumination, where designated, and the weight of the indicator shall not exceed the following:

	4-inch dial	6-inch dial
Height	9-5/8 inches	11-1/2 inches
Width	7 inches	9 inches
Depth	7-5/8 inches	7-5/8 inches
Weight	18 pounds	25 pounds

3.5.2.2.2 Dial marking. - The marking shall be as specified in 3.5.1.2.

3.5.2.2.3 The features applicable to indicators using pointer combination C or D shall be as follows:

3.5.2.2.3.1 Pointer markings. - The outer pointer shall be engraved, or otherwise marked, "P". The inner pointer shall be engraved, or otherwise marked, "S". The markings, "P" and "S", shall be read from top to bottom when the pointer indicates the center of the "STOP" position on the dial.

3.5.2.2.3.2 Terminal markings. - Terminals marked 1MB, 1MBB, 1MB1, 1MB2, and 1MB3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the inner pointer. Terminals marked 2MB, 2MBB, 2MB1, 2MB2, and 2MB3 shall be provided for connection to the type 1F or type 23TR4 synchro receiver associated with the outer pointer. The eleventh and twelfth terminals shall be provided with the double markings 1MB7, 1MB8, and 2MB7, 2MB8, respectively. Audible signals, when used in conjunction with the indicator shall be connected across (a) the terminal marked 1MB7, 1MB8 and either terminal 1MB or terminal 1MBB; and (b) the terminal marked 2MB7, 2MB8 and either terminal 2MB or terminal 2MBB.

Note. - The engine order indicators using pointer combination C or D now active are as follows:

Active types
IC/2UA01
IC/2UD01
IC/2UB01
IC/2UE01

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3.5.2.2.4 The features applicable to indicators using pointer combination E or F shall be as follows:

3.5.2.2.4.1 Pointer markings.- The outer pointer shall be engraved, or otherwise marked, "A". The inner pointer shall be engraved, or otherwise marked, "O". The markings, "A" and "O", shall be read from top to bottom when the pointer indicates the center of the "STOP" position on the dial.

3.5.2.2.4.2 Terminal markings.- Terminals marked MBO, MBBO, MB1, MB2, and MB3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the inner pointer. Terminals marked MBA, MBBA, MB4, MB5, and MB6 shall be provided for connection to the type 1F or type 23TR4 synchro receiver associated with the outer pointer.

Note.- The engine order indicators using pointer combination E or F now active are as follows:

Active types

IC/2UA05
IC/2UD05
IC/2UB05
IC/2UE05

3.5.2.3 Indicator, motor order.- The motor order indicator shall be identical to the indicator specified in 3.5.2.2, using pointer combination C or D (see 3.3.6.2), except for dial markings.

3.5.2.3.1 Dial marking.- Dial markings, lettering, and graduations shall be in accordance with figure 4 of Drawing 9000-S6504-73687.

Note.- The motor order indicators now active are as follows:

Active types

IC/2UA01
IC/2UB01

3.5.2.4 Indicator, plane angle, bow.- The plane angle, bow indicator shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type specified in the contract or order. No provision shall be made for a pedestal mounting. The dial markings and the size and weight shall be as specified in 3.5.1.3.1.1 and 3.5.2.2.1, respectively.

3.5.2.4.1 Pointers.- Pointer combination E (when internal dial illumination is designated) or F (when dial illumination is not designated) shall be used. The outer pointer shall be engraved, or otherwise marked, "BOW". The inner pointer shall be engraved, or otherwise marked, "SET". The markings, "BOW" and "SET", shall be read from left to right when the pointers indicate the zero position on the dial.

3.5.2.4.2 Terminal markings.- Terminals marked 1NB, 1NBB, 1NB1, 1NB2, and 1NB3 shall be provided for connection to the type 1F or 23TR4 synchro receiver associated with the outer pointer. Terminals marked 2NB, 2NBB, 2NB1, 2NB2, and 2NB3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the inner pointer.

Note.- The plane angle bow indicators now active are as follows:

Active types

IC/2UA06
IC/2UB06

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3.5.2.5 Indicator, plane angle, stern.- The plane angle, stern indicator shall be identical to the indicator specified in 3.5.2.4, except for dial, pointer, and terminal markings. The dial markings shall be as specified in 3.5.1.4.1.

3.5.2.5.1 Pointer markings.- The outer pointer shall be engraved, or otherwise marked, "STERN". The inner pointer shall be engraved, or otherwise marked, "SET". The markings, "STERN" and "SET", shall be read from left to right when the pointers indicate the zero position on the dial.

3.5.2.5.2 Terminal markings.- Terminals marked 1NS, 1NSS, 1NS1, 1NS2, and 1NS3 shall be provided for connection to the type 1F or 23TR4 synchro receiver associated with the outer pointer. Terminals 2NS, 2NSS, 2NS1, 2NS2, and 2NS3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the inner pointer.

Note.- The plane angle stern indicators now active are as follows:

Active types

IC/2UA06
IC/2UB06

3.5.2.6 Indicator, propeller order.- The propeller order indicator shall be provided with two dials of the rotating disc type, designated as special dial size by the type. (One cable for audible signals will be furnished (see 3.5.2).)

3.5.2.6.1 Size and weight.- The overall dimensions of the indicator, including the knob for control of the dial illumination, where designated and the weight of the indicator shall not exceed the following:

Bulkhead, panel, and universal mountings

Height	8-1/2 inches
Width	9-1/4 inches
Depth	7 inches
Weight	20 pounds

3.5.2.6.1.1 Pedestal mounting.- The overall dimensions of the indicator provided with a pedestal mounting shall be in accordance with 3.5.2.6.1 and Specification MIL-I-983. The combined weight of the indicator and the associated pedestal shall not exceed 85 pounds.

3.5.2.6.2 Dials.- The two dials shall be arranged so as to provide a range of 00 to 109 revolutions per minute (r.p.m.).

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3.5.2.6.2.1 Dial numerals.- Numerals "0" to "9" inclusive, shall be arranged around the right dial in a counterclockwise direction for increasing values; and numerals "0" to "10" inclusive, shall be arranged around the left dial in a counterclockwise direction for increasing values. The numerals shall not be less than 3/4 inch in height. The arrangement of the numerals around the dials, and the relative location of the two dials, shall be in accordance with figure 1.

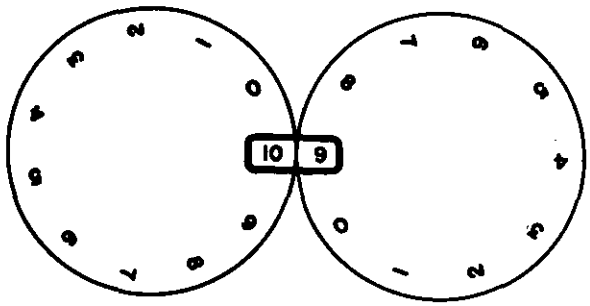


FIGURE 1 — DIAL NUMERAL ARRANGEMENT

3.5.2.6.3 Instrument window.- The indicator shall be provided with a single window. A masking plate shall be incorporated between the instrument window and the rotating disc type dials. The size of the opening in the masking plate shall be such as to display a single numeral on each dial, and exclude display of either adjacent numeral on each of the respective dials. The color of the masking plate shall be the same as the color of the dials. The indicator window opening shall be rectangular in shape, and shall be of such size as to show a 1/4 inch masking plate border around the opening in the masking plate.

3.5.2.6.4 Synchro receivers.- Each rotating disc type dial shall be positioned directly by a type 1F or 23TR4 synchro receiver.

3.5.2.6.5 Audible signal.- A type B3S4 or B3F4 bell, shall be attached to the indicator when a pedestal mounting is designated by the type.

3.5.2.6.6 Terminal markings.- Terminals marked M, MM, M1, M2, and M3 shall be provided for connection to the synchro receiver associated with the left dial. Terminals marked M, MM, M4, M5, and M6 shall be provided for connection to the synchro receiver associated with the right dial. Terminals marked M shall be connected together. Terminals marked MM shall be connected together. Terminal

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markings M7 and M8 shall be provided for the eleventh and twelfth terminals. An audible signal, when used in conjunction with the indicator shall be connected across terminals MM and M7. The signal pushbutton incorporated in the indicator shall be connected across terminals M and M8.

Note.- The propeller order indicators now active are as follows:

Active types

IC/2UF03
IC/2SF04

3.5.2.7 Indicator, rudder angle.- The rudder angle indicator shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type. The indicator shall normally be provided with pointer combination C or D (depending on whether or not internal dial illumination is designated) unless pointer combination E or F is designated.

3.5.2.7.1 Size and weight.- The size and weight shall conform to 3.5.2.2.1 for the indicator when bulkhead, panel, or universal mounting is designated by the type. The overall dimensions of the indicator provided with a pedestal mounting shall be in accordance with 3.5.2.2.1 and Specification MIL-I-983. The combined weight of the indicator and the associated pedestal shall not exceed the following:

4-inch dial - 85 pounds
6-inch dial - 95 pounds

3.5.2.7.2 Dial marking.- The dial marking shall conform to 3.5.1.6.2.

3.5.2.7.3 The features applicable to indicators using pointer combinations C or D shall be as follows:

3.5.2.7.3.1 Pointer markings.- The outer pointer shall be engraved, or otherwise marked, "P". The inner pointer shall be engraved, or otherwise marked "S". The markings, "P" and "S", shall be read from top to bottom when the pointer indicates the zero position on the dial.

3.5.2.7.3.2 Terminal markings.- Terminals marked 1N, 1NN, 1N1, 1N2, and 1N3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the inner pointer. Terminals marked 2N, 2NN, 2N1, 2N2, and 2N3 shall be provided for connection to the type 1F or 23TR4 synchro receiver associated with the outer pointer.

Note.- The rudder angle indicators using pointer combination C or D now active are as follows:

Active types

IC/2UA11
IC/2UD11
IC/2UB11
IC/2UE11

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3.5.2.7.4 The features applicable to indicators using pointer combination E or F shall be as follows:

3.5.2.7.4.1 Pointer markings.- The outer pointer shall be engraved, or otherwise marked, "RUDDER". The inner pointer shall be engraved, or otherwise marked, "SET". The markings, "RUDDER" and "SET", shall be read from top to bottom when the pointer indicates the zero position on the dial.

3.5.2.7.4.2 Terminal markings.- Terminals marked 1N, 1NN, 1N1, 1N2, and 1N3 shall be provided for connection to the type 1F or 23TR4 synchro receiver associated with the outer pointer. Terminals marked 5N, 5NN, 5N1, 5N2, and 5N3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the inner pointer.

Note.- The rudder angle indicators using pointer combination E or F are as follows:

Active types

IC/2UA06
IC/2UB06

3.5.2.8 Indicator, rudder angle-order.- The rudder angle-order indicator shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type. The pointer combination shall be E or F (see table V). The dial markings and the size and weight shall be as specified in 3.5.1.6.2 and 3.5.2.7.1, respectively. (One cable for audible signals will be furnished (see 3.5.1).)

3.5.2.8.1 Pointer markings.- The outer pointer shall be engraved, or otherwise marked, "ANGLE". The inner pointer shall be engraved, or otherwise marked, "ORDER". The markings, "ANGLE" and "ORDER", shall be read from top to bottom when the pointer indicates the zero position on the dial.

3.5.2.8.2 Audible signal.- A type B2S4 or B2F4 bell, shall be attached to the indicator when a pedestal mounting is designated by the type.

3.5.2.8.3 Terminal markings.- Terminals marked N, NN, N1, N2, and N3 shall be provided for connection to the type 1F or 23TR4 synchro receiver associated with the outer pointer. Terminals L, LL, L1, L2, and L3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the inner pointer. A terminal marked L4 shall be provided for the audible signal. The audible signal, when used with the indicator, shall be connected across terminals LL and L4.

Note.- The rudder angle-order indicators now active are as follows:

Active types

IC/2UD01
IC/2UE01

3.5.3 Indicators, triple.- The triple indicators shall provide three indications; and shall incorporate either a single dial and three concentric pointers, or three separate dials of the rotating disc type.

3.5.3.1 Indicator, propeller order.- The propeller order indicator shall be provided with three dials of the rotating disc type, designated as special size by the type specified in the contract or order. The synchro receivers and audible signal shall conform to 3.5.2.6.4 and 3.5.2.6.5, respectively. (One cable for audible signal will be furnished (see 3.5.1).)

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3.5.3.1.1 Size and weight.- The overall dimensions of the indicator, and the weight of the indicator shall not exceed the following:

Bulkhead, panel, and universal mountings

Height	8-1/2 inches
Width	8-1/2 inches
Depth	7-1/2 inches
Weight	22 pounds

3.5.3.1.1.1 Pedestal mounting.- The overall dimensions of the indicator provided with a pedestal mounting shall be in accordance with 3.5.3.1.1 and Specification MIL-I-983. The combined weight of the indicator and the associated pedestal shall not exceed 90 pounds.

3.5.3.1.2 Dials.- The three dials shall be arranged so as to provide a range of 000 to 999 r.p.m.

3.5.3.1.2.1 Dial numerals.- Numerals "0" to "9" inclusive, shall be arranged around each dial in a counterclockwise direction for increasing values. The arrangement of the numerals around the "hundreds" (left) dial and the "units" (right) dial shall be in accordance with figure 11 of Drawing 9000-S6504-73687. The arrangement of the numerals around the "tens" (center) dial shall be in accordance with figure 12 of Drawing 9000-S6504-73687. The numerals shall be not less than 3/4 inch in height. The dials shall be located in the indicator so that the numerals at the bottom of the "hundreds" and "units" dials are in a horizontal line with the numeral at the top of the "tens" dial.

3.5.3.1.3 Instrument windows.- The indicator shall be provided with three windows, one for each dial. A masking plate shall be incorporated between each window and the associated disc type dial. The size of the opening in each masking plate shall be such as to display a single numeral on the associated dial, and exclude display of either adjacent numeral on the dial. The color of the masking plates shall be the same as the color of the dials. The indicator window openings shall each be rectangular in shape, and shall be of such size as to show a 1/4 inch masking plate border around the opening in the masking plate.

3.5.3.1.4 Terminal markings.- Terminals marked M, MM, M1, M2, and M3 shall be provided for connection to the synchro receiver associated with the "hundreds" dial. Terminals marked M, MM, M4, M5, and M6 shall be provided for connection to the secondary winding of the synchro receiver associated with the "tens" dial. Terminals marked M, MM, M7, M8, and M9 shall be provided for connection to the secondary winding of the synchro receiver associated with the "units" dial. Terminals marked M shall be connected together. Terminals marked MM shall be connected together. A terminal marked M10 shall be provided for an audible signal. An audible signal, when used in conjunction with the indicator, shall be connected across terminals MM and M10.

Note.- Cable for connection to ship's wiring - An MSCA-14 cable (see Spec. MIL-C-2194) or an MHFA-14 cable (see Spec. MIL-C-915) will be brought into the bottom of the instrument for connection of ship's wiring. The propeller order indicators now active are as follows:

Active types

IC/3UF01

3.5.3.2 Indicator, rudder angle-order.- The rudder angle-order indicator shall be provided with a single dial and three concentric pointers. A 6-inch dial shall be provided. Except as specified herein, the indicator shall embody the characteristics of the rudder angle indicator (double) specified in 3.5.2.7, and the characteristics of the rudder order indicator specified in 3.5.1. Pointer combination G or H (see 3.3.6.2) shall be provided. (*One cable for audible signal will be furnished (see 3.5.1).*)

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3.5.3.2.1 Size and weight.- The overall dimensions of the indicator, including all knobs, and the weight of the indicator-transmitter shall not exceed the following:

Bulkhead, panel, and universal mountings

Height	12 inches
Width	10 inches
Depth	8 inches
Weight	40 pounds

3.5.3.2.1.1 Pedestal mounting.- The overall dimensions of the indicator provided with a pedestal mounting shall be in accordance with 3.5.3.2.1 and Specification MIL-I-983. The combined weight of the indicator and the associated pedestal shall not exceed 105 pounds.

3.5.3.2.2 Pointers.- The outer and middle pointers of the indicator shall be in accordance with the requirements for the outer and inner pointers, respectively, specified in 3.5.2. The inner pointer shall be marked in accordance with 3.5.1.7.1. The inner pointer shall be positioned by a type 3F or 31TR4 synchro receiver. The inner pointer shall be geared to the associated synchro receiver, and the motion transferred between the synchro receiver gear and the inner pointer gear shall be in a 1 to 1 ratio.

3.5.3.2.3 Terminal markings.- Terminals marked 1N, 1NN, 1N1, 1N2, 1N3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the middle pointer. Terminals marked 2N, 2NN, 2N1, 2N2, 2N3 shall be provided for connection to the type 1F or 23TR4 synchro receiver associated with the outer pointer. Terminals marked L, LL, L1, L2, L3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the inner pointer. A terminal marked L4 shall be provided for the audible signal. The audible signal, when used with the indicator, shall be connected across terminals LL and L4.

Note.- Cable for connection to ship's wiring - A type MSCA-19 cable (see Spec. MIL-C-2194) or a type MHFA-19 cable (see Spec. MIL-C-915) will be brought into the bottom of the instrument for connection of ship's wiring. The rudder angle-order indicators now active are as follows:

Active types

IC/3UE01

3.5.3.3 Indicator, rudder angle.- The rudder angle indicator shall be provided with a single dial and three concentric pointers. A 6-inch dial shall be provided. The dial markings shall conform to 3.5.1.6.2. Pointers combination I or J (see table V) shall be provided. The size and weight shall conform to 3.5.3.2.1 and 3.5.3.2.1.1. Operation of the pointers shall be in accordance with 3.5.3.2. (Cable for connection to ship's wiring will be as shown in note following 3.5.3.2.3.)

3.5.3.3.1 Pointer markings.- The outer pointer shall be engraved or otherwise marked "P". The middle pointer shall be engraved or otherwise marked "C". The inner pointer shall be engraved or otherwise marked "S". The markings "P", "C", and "S" shall be read from top to bottom when the pointer indicates the "0" position on the dial.

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3.5.3.3.2 Terminal markings.- Terminals marked 1N, 1NN, 1N1, 1N2, and 1N3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the inner pointer. Terminals marked 2N, 2NN, 2N1, 2N2, and 2N3 shall be provided for connection to the type 1F or 23TR4 synchro receiver associated with the outer pointer. Terminals marked 3N, 3NN, 3N1, 3N2, and 3N3 shall be provided for connection to the type 3F or 31TR4 synchro receiver associated with the middle pointer.

Note.- The rudder angle indicators now active are as follows:

Active types
IC/3UB11
IC/3UE11

3.6 Transmitters.- Type 5G or 37TX4 synchro transmitters shall be used in all transmitters. Transmitters shall be designated as single or double as specified herein.

3.6.1 Transmitters, single.- Each single transmitter shall receive a single mechanical input, and transmit a single synchro signal. Display of the order or information transmitted, where required, shall be as specified herein. When the type designation specifies that dial illumination shall be incorporated in the indicator, the illumination power supply shall be taken from the power supply of the synchro transmitter. Unless otherwise specified herein, the terminal strip in each single transmitter shall be provided with 6 terminals. (Unless otherwise specified herein, a type MSCA-7 cable (see Spec. MIL-C-2194) or a type MHFA-7 cable (see Spec. MIL-C-915) will be brought into the single transmitters for connection of ship's wiring.) For the single transmitters specified in 3.6.1.1, 3.6.1.2, and 3.6.1.4, provision shall be made for the entry of ship's wiring through one of the instrument surfaces adjacent to the surface which contains the operating shaft, but exclusive of either the mounting surface or the instrument cover. (A DSGA-3 cable (see Spec. MIL-C-2194) or a DHFA-3 cable (see Spec. MIL-C-915) will be brought into the top (see 3.3.8) or bottom (see 3.3.9) of bulkhead, panel, and universal mounted instruments for connection to the audible signal associated with the applicable instrument.)

3.6.1.1 Transmitter, plane angle, bow.- The plane angle, bow transmitter shall be designed for bulkhead mounting. The enclosure for the transmitter shall be of cast bronze in accordance with Specification 46B24. The transmitter shall not be provided with plug and jack connections for the internal assembly. The terminal markings shall conform to 3.5.1.3.4.

3.6.1.1.1 Size and weight.- The overall dimensions, exclusive of the operating shaft external to the enclosure and the weight shall not exceed the following:

	5G synchro	6G synchro
Height	9-3/4 inches	10 inches
Width	9-1/2 inches	9-3/4 inches
Depth	5-1/2 inches	6 inches
Weight	25 pounds	35 pounds

3.6.1.1.2 Positioning the synchro transmitter.- The synchro transmitter shall be positioned through gearing by the rotation of the operating shaft of the transmitter.

3.6.1.1.2.1 Ratio of angular movement.- The ratio of the angular movement of the synchro transmitter to the angular movement of the operating shaft shall be 2 to 1.

3.6.1.1.2.2 Gearing.- Sector gears, quadrant gears and bevel gears shall not be used in the transmitter.

3.6.1.1.2.3 Interchanging of gears.- The design of the transmitter shall be such that the ratio of angular movement can be changed from 2:1 to 1:2 by interchanging the gears.

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3.6.1.1.2.4 Operating shaft.- The operating shaft external to the enclosure shall be 0.625 inch in diameter. The length of the shaft external to the enclosure shall not be less than 1 inch beyond the outermost protrusion of the enclosure. The length of the shaft shall be such as to permit rotation of the lever (see 3.6.1.1.3) through 360 degrees. A keyway shall be provided in the operating shaft external to the enclosure; and an associated key shall also be provided with the transmitter.

3.6.1.1.3 Lever.- A lever shall be attached to the operating shaft when a lever is designated by the type.

3.6.1.1.3.1 Length of lever.- The lever shall provide a length of 12 inches between the axis of the transmitter operating shaft and the axis of rotation of the coupling attached to the end of the lever.

3.6.1.1.3.2 Couplings.- A coupling shall be attached to the end of the lever to permit connection to a linkage between the lever and the source of mechanical input to the lever. An additional and similar coupling shall be provided with the lever. (This additional coupling will permit attachment to the source of mechanical input to the lever, and will provide the connection for the linkage between the lever and the source of mechanical input.) Each coupling shall contain a 1/2 inch iron pipe size (i.p.s.) tap to receive the threaded ends of the linkage.

Note.- The plane angle bow transmitters now active are as follows:

Active types
IC/4BG21
IC/4BG20

3.6.1.2 Transmitter, plane angle, stern.- The plane angle, stern-transmitter shall be identical to the transmitter specified in 3.6.1.1, except that the terminal markings shall conform to 3.5.1.4.3.

Note.- The plane angle stern transmitters now active are as follows:

Active types
IC/4BG21
IC/4BG20

3.6.1.3 Transmitter, propeller order.- The propeller order transmitter shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type. The dial marking and terminals shall be as specified in 3.5.1.5.2 and 3.5.1.5.4, respectively. (Cables for connection to ship's wiring will be as shown in note following 3.5.1.5.5.)

3.6.1.3.1 Size and weight.- The overall dimensions of the transmitter, including the operating knob, and the weight of the transmitter, shall not exceed the following:

Bulkhead, panel, and universal mountings		
	4 inch dial	6 inch dial
Height	10 inches	12 inches
Width	7 inches	9 inches
Depth	9-1/4 inches	9-1/4 inches
Weight	25 pounds	35 pounds

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3.6.1.3.1.1 Pedestal mounting. - The overall dimensions of the indicator provided with a pedestal mounting shall be in accordance with 3.6.1.3.1 and Specification MIL-I-983. The combined weight of the transmitter and the associated pedestal shall not exceed the following:

4 inch dial - 90 pounds
6 inch dial - 100 pounds

3.6.1.3.2 Operating knob. - The synchro transmitter and instrument pointer shall be positioned by means of an operating knob. Clockwise rotation of the operating knob shall result in clockwise rotation of the instrument pointer.

3.6.1.3.2.1 Location. - The operating knob shall be located directly below the instrument window opening.

3.6.1.3.2.2 Ratio of angular movement. - The ratio of the angular movement of the operating knob to the angular movement of the synchro transmitter, and to the angular movement of the instrument pointer, shall be 1:1.

3.6.1.3.2.3 Operating mechanism. - The operating mechanism shall be provided with a suitable braking feature (friction brake) to ensure stability of positioning.

3.6.1.3.3 Audible signal. - A type Z1S4 or Z1F4 buzzer, shall be attached to the transmitter when a pedestal mounting is designated by the type.

3.6.1.3.4 Terminal markings. - Terminals marked M, MM, M1, M2, and M3 shall be provided for connection to the synchro transmitter. Terminal markings M7 and M8 shall be provided for the sixth and seventh terminals, respectively. An audible signal, when used in conjunction with the transmitter, shall be connected across terminals MM and M8. The signal push button incorporated in the transmitter shall be connected across terminals M and M7.

Note. - The propeller order transmitters now active are as follows:

Active types

IC/4UB27
IC/4UA27
IC/4SB28
IC/4SA28

3.6.1.4 Transmitter, rudder angle. - The rudder angle transmitter shall be identical to the transmitter specified in 3.6.1.1, except for ratio of angular movement, interchanging of gears and terminal markings.

3.6.1.4.1 Ratio of angular movement. - The ratio of the angular movement of the synchro transmitter to the angular movement of the operating shaft shall be 4:1.

3.6.1.4.1.1 Interchange of gears. - The design of the transmitter shall be such that the ratio of angular movement can be changed from 4:1, to 1:4 by interchanging the gears.

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3.6.1.4.2 Terminal markings.- Terminal markings N, NN, N1, N2, and N3 shall be provided.

Note.- The rudder angle transmitters now active are as follows:

Active types

IC/4BG20
IC/4BG21
IC/4BG31
IC/4BG32

3.6.1.5 Transmitter, rudder order.- The rudder order transmitter shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type. The dial marking, pointer marking and the size and weight shall be as specified in 3.5.1.6.2, 3.5.1.7.1 and 3.6.1.3.1, respectively.

3.6.1.5.1 Operating knob.- The synchro transmitter and instrument pointer shall be positioned by means of an operating knob. Clockwise rotation of the operating knob shall result in counterclockwise rotation of the instrument pointer. The location, ratio of angular movement and the operating mechanism shall be as specified in 3.6.1.3.2.1, 3.6.1.3.2.2 and 3.6.1.3.2.3, respectively.

3.6.1.5.2 Pilot light.- The power supply for the pilot light shall be taken from the power supply of the synchro transmitter.

3.6.1.5.3 Terminal markings.- Terminals marked L, LL, L1, L2, and L3 shall be provided for connection to the synchro transmitter. Terminal marking L4 shall be provided for the sixth terminal. The signal push button incorporated in the transmitter shall be connected across terminals L and L4.

Note.- The rudder order transmitters now active are as follows:

Active types

IC/4SB22

3.6.2 Transmitters, double.- Each double transmitter shall receive two mechanical inputs, and transmit two separate synchro signals. Display of the orders or information transmitted shall be as specified herein. When the type designation specifies that dial illumination shall be incorporated in the transmitter, the illumination power supply shall be taken from the power supply of one of the synchro transmitters as specified herein. Unless otherwise specified herein, each double transmitter shall be provided with two terminal strips of 6 terminals each. (Cables for connection to ships wiring will be as shown in note following 3.5.3.1.4. One cable for audible signal will be furnished (see 3.5.1).)

3.6.2.1 Transmitter, course-steering order.- The course steering order transmitter shall be provided with a 6-inch dial. The dial marking shall be as specified in 3.5.2.1.2.

3.6.2.1.1 Size and weight.- The overall dimensions of the transmitter, including all knobs, and the weight of the transmitter shall not exceed the following:

Bulkhead, panel, and universal mountings

Height	12 inches
Width	10 inches
Depth	9-1/4 inches
Weight	40 pounds

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3.6.2.1.1.1 Pedestal mounting. - The overall dimensions of the transmitter provided with a pedestal mounting shall be in accordance with 3.6.2.1.1 and Specification MIL-I-983. The combined weight of the indicator and the associated pedestal shall not exceed 105 pounds.

3.6.2.1.2 Course orders. - The synchro transmitter and the associated course dial ring shall be positioned by means of an operating knob. Clockwise rotation of the operating knob shall result in counter-clockwise rotation of the course dial ring (for increasing values of desired course order indications).

3.6.2.1.2.1 Location of operating knob. - The operating knob shall be located below and to the right of the instrument window opening.

3.6.2.1.2.2 Ratio of angular movement. - The ratio of the angular movement of the operating knob to angular movement of the synchro transmitter, and to the angular movement of the course order dial ring, shall be 2:1.

3.6.2.1.2.3 Operating mechanism. - The operating mechanism for the course order transmitter shall be provided with a suitable braking feature (friction brake) to ensure stability of positioning.

3.6.2.1.3 Steering orders. - The synchro transmitter and the associated pointer shall be positioned by means of an operating knob. Clockwise rotation of the operating knob shall result in clockwise rotation of the pointer.

3.6.2.1.3.1 Location of operating knob. - The operating knob shall be located below and to the left of the instrument window opening.

3.6.2.1.3.2 Ratio of angular movement. - The ratio of the angular movement of the operating knob to the angular movement of the synchro transmitter, and to the angular movement of the steering order pointer, shall be 1:1.

3.6.2.1.3.3 Operating mechanism. - The operating shaft of the steering order transmitter shall be fitted with a detent wheel to ensure stability of positioning, and to operate electrical contacts in an audible signal circuit. The detent wheel and the signal contacts shall ensure proper operation (signal contacts shall close and remain closed for a sufficient period of time) of the remote audible signal when the steering order transmitter operating knob is rotated rapidly from one position to another. The detent wheel shall be provided with a stop at a point corresponding to the center of the space between the two "HARD OVER" orders shown on the steering order section of the instrument dial.

3.6.2.1.4 Audible signal. - A type B1S4 or B1F4 bell, shall be attached to the transmitter when a pedestal mounting is designated by the type.

3.6.2.1.5 Terminal markings. - Terminals marked LA/C, LAA/C, LA1, LA2, and LA3 shall be provided for connection to the synchro transmitter associated with the course order portion of the instrument. Terminals marked LA/S, LAA/S, LA4, LA5, and LA6 shall be provided for connection to the synchro transmitter associated with the steering order portion of the instrument. Terminals marked LA7 and LA8 shall be provided for connection to the audible signal and signal push button (and signal contacts specified in 3.6.2.1.3.3), respectively. The signal push button incorporated in the transmitter shall be connected in parallel with the signal contacts specified in 3.6.2.1.3.3, and these signal devices shall then be connected across terminals LA and LA8. An audible signal, when used in conjunction with the transmitter, shall be connected across terminals LAA and LA7.

Note. - The course-steering order now active are as follows:

Active types

IC/5SB24
IC/5UB23

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3.6.2.2 Transmitter, propeller order. - The propeller order transmitter shall be provided with two dials of the rotating disc type, designated as special dial size by the type. The dials, dial numerals and instrument window shall be as specified in 3.5.2.6.2, 3.5.2.6.2.1 and 3.5.2.6.3, respectively.

3.6.2.2.1 Size and weight. - The overall dimensions of the transmitter, including all knobs, and the weight of the transmitter shall not exceed the following:

Bulkhead, panel, and universal mountings

Height	12 inches
Width	10 inches
Depth	9-1/4 inches
Weight	40 pounds

3.6.2.2.1.1 Pedestal mounting. - The overall dimensions of the transmitter provided with a pedestal mounting shall be in accordance with 3.6.2.2.1 and Specification MIL-I-983. The combined weight of the transmitter and the associated pedestal shall not exceed 105 pounds.

3.6.2.2.2 Propeller orders. - Each synchro transmitter and the associated dial shall be positioned by means of separate operating knobs. Clockwise rotation of each knob shall result in clockwise rotation of the associated dial.

3.6.2.2.2.1 Location of operating knobs. - The operating knob for the right dial shall be located to the right of the instrument window opening. The operating knob for the left dial shall be located to the left of the instrument window opening.

3.6.2.2.2.2 Ratio of angular movement. - The ratio of the angular movement of each operating knob to the angular movement of the associated synchro transmitter, and to the associated dial, shall be 1:1.

3.6.2.2.2.3 Operating mechanism. - Each operating shaft shall be fitted with a detent wheel to ensure stability of positioning, and to operate electrical contacts in an audible signal circuit. Each detent wheel and the associated signal contacts shall ensure proper operation (signal contacts shall close and remain closed for a sufficient period of time) of the remote audible signal when the operating knobs are rotated rapidly from one position to another.

3.6.2.2.3 Audible signal. - A type Z1S4 or Z1F4 buzzer shall be attached to the transmitter when a pedestal mounting is designated by the type.

3.6.2.2.4 Terminal markings. - Terminals marked M, MM, M1, M2, and M3 shall be provided for connection to the synchro transmitter associated with the left dial. Terminals marked M, MM, M3, M4, and M5 shall be provided for connection to the synchro transmitter associated with the right dial. Terminals marked M shall be connected together. Terminals marked MM shall be connected together. Terminals marked M7 and M8 shall be provided for the eleventh and twelfth terminals. An audible signal, when used in conjunction with the transmitter, shall be connected across terminals MM and M8. The signal push button incorporated in the transmitter shall be connected in parallel with the signal contacts specified in 3.6.2.2.2.3, and these signal devices shall then be connected across terminals M and M7.

Note. - The propeller order transmitters now active are as follows:

Active types

IC/5SC26
IC/5UC25

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3.7 Indicator-transmitters. - Each indicator-transmitter shall contain, in a single enclosure, the mechanism necessary to receive and indicate the order(s), or information data received from remote transmitters; and the mechanism necessary to receive the mechanical input(s), to transmit the associated synchro signal(s), and display the order(s), or information transmitted. Indicator-transmitters shall be designated as single, single (special), double and triple. Type 1F or 23TR4 synchro receivers shall be used for all applications where an indicator pointer is positioned directly by the synchro receiver. Unless otherwise specified in the contract or order, type 3F or 31TR4 synchro receivers shall be used for all applications where an indicator pointer is positioned through gearing. A type 5G or 37TX4 synchro transmitter shall be used for all transmitter applications unless a type 6G synchro is specified in the contract or order. All transmitter synchros shall be positioned by the use of operating knobs or handles.

3.7.1 Indicator-transmitters, single. - Each single indicator-transmitter shall receive a single synchro signal from a remote synchro transmitter, and shall indicate or display the order or information data received. Each single indicator-transmitter shall also receive a single mechanical input, and shall transmit a single synchro signal corresponding to the mechanical input, and shall display the order transmitted. Unless otherwise specified herein, each single indicator-transmitter shall be provided with two concentric pointers. When the type designation specifies that dial illumination shall be incorporated in the indicator-transmitter, the illumination power supply shall be taken from the power supply of the type 1F or 23TR4 synchro receiver. Unless otherwise specified herein, each single indicator-transmitter shall be provided with two terminal strips of 6 terminals each. (Cables for connection to ships wiring will be as shown in note following 3.5.3.1.4.)

3.7.1.1 Indicator-transmitter, anchor order. - The anchor order indicator-transmitter shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type. (One cable for audible signal will be furnished (see 3.5.1).)

3.7.1.1.1 Size and weight. - The overall dimensions of the indicator-transmitter, including all knobs, and the weight of the indicator-transmitter shall not exceed the following:

Bulkhead, panel, and universal mountings

Height	12 inches
Width	9 inches
Depth	9-1/4 inches
Weight	35 pounds

3.7.1.1.1.1 Pedestal mounting. - The overall dimensions of the indicator-transmitter provided with a pedestal mounting shall be in accordance with 3.7.1.1.1 and Specification MIL-I-983. The combined weight of the indicator-transmitter and the associated pedestal shall not exceed the following:

4-inch dial	95 pounds
6-inch dial	100 pounds

3.7.1.1.2 Dial marking. - Dial markings, lettering, and graduations shall be in accordance with figure 6 of Drawing 9000-S6504-73687.

3.7.1.1.3 Pointers. - Pointer combination E (when internal dial illumination is specified) or F shall be installed (see table V).

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3.7.1.1.3.1 Pointer markings. - Pointer markings shall be either of the combinations specified in table VI.

Table VI - Pointer marking combinations.

Combination	Pointer marking	
	Inner	Outer
1	O	A
2	A	O

The markings shall read from top to bottom when the pointer indicates the center of the "OFF" position on the dial.

3.7.1.1.4 Synchros. - The outer pointer shall be positioned directly by a type 1F or 23TR4 synchro receiver. A type 5G or 37TX4 synchro transmitter shall be used for transmitting the order, and shall be positioned in association with the inner pointer.

3.7.1.1.5 Operating knob. - The synchro transmitter and associated inner pointer shall be positioned by means of an operating knob. Clockwise rotation of the operating knob shall result in clockwise rotation of the inner pointer. The location and the ratio of angular movement shall be as specified in 3.6.1.3.2.1 and 3.6.1.3.2.2, respectively.

3.7.1.1.5.1 Operating mechanism. - The operating shaft shall be fitted with a detent wheel to ensure stability of positioning, and to operate electrical contacts in an audible signal circuit. The detent wheel and the signal contacts shall ensure proper operation (signal contacts shall close and remain closed for a sufficient period of time) of the remote audible signal when the operating knob is rotated rapidly from one position to another.

3.7.1.1.6 Pilot light. - The power supply for the pilot light shall be taken from the power supply of the synchro transmitter.

3.7.1.1.7 Audible signals. - When a pedestal mounting is designated by the type for the indicator-transmitter a bell shall be attached to the indicator-transmitter in accordance with the following:

3.7.1.1.7.1 Indicator-transmitter with pointer combination 1. - A type B1S4 or B1F4 bell shall be attached.

3.7.1.1.7.2 Indicator-transmitter with pointer combination 2. - A type B2S4 or B2F4 bell shall be attached.

3.7.1.1.8 Terminal markings. -

3.7.1.1.8.1 Indicator-transmitter with pointer combination 1. - Terminals marked HB-T, HBB-T, HB1, HB2, and HB3 shall be provided for connection to the synchro transmitter. Terminals marked HB-I, HBB-I, HB4, HB5, and HB6, shall be provided for connection to the synchro receiver. Terminals marked HB7 and HB8 shall be provided for connection to the signal push button (and signal contacts specified in 3.7.1.1.5.1) and the audible signal, respectively. The signal push button incorporated in the indicator-transmitter shall be connected in parallel with the signal contacts specified in 3.7.1.1.5.1, and the signal devices shall then be connected across terminals HB-T and HB7. An audible signal, when used in conjunction with the indicator-transmitter, shall be connected across terminals HBB-I and HB8.

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3.7.1.1.8.2 Indicator-transmitter with pointer combination 2. - Terminals marked HB-T, HBB-T, HB4, HB5, and HB6 shall be provided for connection to the synchro transmitter. Terminals marked HB-I, HBB-I, HB1, HB2, and HB3 shall be provided for connection to the synchro receiver. Terminals marked HB7 and HB8 shall be provided for connection to the audible signal and signal push button (and signal contacts specified in 3.7.1.1.5.1), respectively. The signal push button shall be connected in parallel with the signal contacts specified in 3.7.1.1.5.1, and the signal devices shall then be connected across terminals HB-T and HB8. An audible signal, when used in conjunction with the indicator-transmitter, shall be connected across terminals HBB-I and HB7.

Note. - The anchor order indicator-transmitters now active are as follows:

Active types

Indicator-transmitters with pointer combination 1

IC/7UB42
IC/7UA42

Indicator-transmitters with pointer combination 2

IC/7UB42
IC/7UA42
IC/7UE42
IC/7UD42

3.7.1.2 Indicator-transmitter, engine order. - The engine order indicator-transmitter shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type. The size and weight, pointers, synchros, operating knob, pilot light and audible signals shall be as specified in 3.7.1.1.1, 3.7.1.1.3, 3.7.1.1.4, 3.7.1.1.5, 3.7.1.1.6 and 3.7.1.1.7, respectively. (One cable for audible signal will be furnished (see 3.5.1).)

3.7.1.2.1 Dial markings. - Dial markings, lettering, and graduations shall be in accordance with figure 1 of Drawing 9000-S6504-73687.

3.7.1.2.2 Detent wheel. - In addition to the requirements specified in 3.7.1.1.5.1, the detent wheel shall be provided with a stop at a point corresponding to the center of the dial space diametrically opposite to the order "1/3 AHEAD".

3.7.1.2.3 Indicator-transmitter with pointer combination 1 (see 3.7.1.1.3.1). -

3.7.1.2.3.1 Terminal markings. - Terminals marked MB-T, MBB-T, MB1, MB2, and MB3 shall be provided for connection to the synchro transmitter. Terminals marked MB-I, MBB-I, MB4, MB5, and MB6 shall be provided for connection to the synchro receiver. Terminals marked MB7 and MB8 shall be provided for connection to the signal push button (and signal contacts specified in 3.7.1.1.5.1) and the audible signal, respectively. The signal push button incorporated in the indicator-transmitter shall be connected in parallel with the signal contacts specified herein, and the signal devices shall then be connected across terminals MB-T and MB-7. An audible signal, when used in conjunction with the indicator-transmitter, shall be connected across terminals MBB-I and MB8.

Note. - The engine order indicator-transmitters with pointer combination 1 now active are as follows:

Active types

IC/7UB42
IC/7SB43
IC/7UA42
IC/7SA43

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3.7.1.2.4 Indicator-transmitter with pointer combination 2 (see 3.7.1.1.3.1). -

3.7.1.2.4.1 Direction contact switches. - The indicator-transmitter shall incorporate contact switches in the "AHEAD" and "BACK" positions of the operating shaft. The switches are for use in conjunction with an engine direction warning signal system. One contact switch shall be closed automatically whenever the transmitter portion of the instrument is in the "AHEAD" position. The other contact switch shall be closed automatically whenever the transmitter portion of the instrument is in the "BACK" position.

3.7.1.2.4.2 Additional terminal strip. - An additional terminal strip, containing 2 terminals, shall be incorporated to provide terminals for connection to the direction contact switches specified in 3.7.1.2.4.1.

3.7.1.2.4.3 Terminal markings. - Terminals marked MB-T, MBB-T, MB4, MB5, and MB6 shall be provided for connection to the synchro transmitter. Terminals marked MB-I, MBB-I, MB1, MB2, and MB3 shall be provided for connection to the synchro receiver. Terminals marked MB7 and MB8 shall be provided for connection to the audible signal and signal push button (and signal contacts specified in 3.7.1.1.5.1), respectively. The signal push button shall be connected in parallel with the signal contacts specified immediately above, and these signal devices shall then be connected across terminals MB-T and MB8. An audible signal, when used in conjunction with the indicator-transmitter, shall be connected across terminals MBB-I and MB7. Terminals marked MB9 and MB10 shall be provided for connection to the direction contact switches specified in 3.7.1.2.4.1. The contact switch which is closed when the transmitter is in the "AHEAD" position shall be connected across terminals MB-I and MB9. The contact switch which is closed when the transmitter is in the "BACK" position shall be connected across terminals MB-I and MB10.

3.7.1.2.4.4 Additional cable entrance. - (A FSGA-3 cable (see Spec. MIL-C-2194) or FHFA-3 cable (see Spec. MIL-C-915) will be used for connection to the direction contact switch circuit.) Provision shall be made for cable entrance in the top of cast enclosures (see 3.3.8) and in the bottom of sheet metal enclosures (see 3.3.9).

Note. - The engine order indicator-transmitter with pointer combination 2 now active are as follows:

Active types

IC/7UE44
IC/7UD44

3.7.1.3 Indicator-transmitter, motor order. - The motor order indicator-transmitter shall be identical to the indicator-transmitter specified in 3.7.1.2, except for dial markings and the additional detent positions required for the increase in active segments of the dial.

3.7.1.3.1 Dial marking. - Dial markings, lettering, and graduations shall be in accordance with figure 4 of Drawing 9000-S6504-73687.

Note. - The motor order indicator-transmitters now active are as follows:

Active types

IC/7UB42
IC/7UA42
IC/7UE44
IC/7UD44

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3.7.1.4 Indicator-transmitter, rudder angle-order. - The indicator-transmitter shall be provided with a 4-inch dial or a 6-inch dial, as designated by the type. The dial markings, size and weight and the pilot light shall be as specified in 3.5.1.6.2, 3.7.1.1.1, and 3.7.1.1.6, respectively. Pointer combination E (when internal dial illumination is specified) or F shall be installed (see table V).

3.7.1.4.1 Outer pointer. - The outer pointer shall be positioned directly by the synchro receiver (type 1F or 23TR4) and shall be worked in accordance with 3.5.1.6.3.

3.7.1.4.2 Inner pointer. - The inner pointer and associated synchro transmitter (type 5G or 37TX4) shall be positioned by means of an operating knob. The pointer marking shall be in accordance with 3.5.1.7.1.

3.7.1.4.3 Operating knob. - The synchro transmitter and associated inner pointer shall be positioned by means of an operating knob. Clockwise rotation of the operating knob shall result in counterclockwise rotation of the associated inner pointer. The location, ratio of angular movement and operating mechanism shall be as specified in 3.6.1.3.2.1, 3.6.1.3.2.2 and 3.6.1.3.2.3, respectively.

3.7.1.4.4 Terminal markings. - Terminals marked N, NN, N1, N2, and N3 shall be provided for connection to the synchro receiver. Terminals marked L, LL, L1, L2, and L3 shall be provided for connection to the synchro transmitter. Terminal marking L4 shall be provided for the eleventh terminal. The signal push button incorporated in the indicator-transmitter shall be connected across terminals L and L4.

Note. - The rudder angle-order indicator-transmitters now active are as follows:

Active types

IC/7SB48
IC/7UE48

3.7.2 Indicator-transmitters, single (special). - Each single (special) indicator-transmitter shall receive a single mechanical input, shall transmit a single synchro signal corresponding to the mechanical input, and shall display the order transmitted. Each single (special) indicator-transmitter shall receive two separate synchro signals from separate remote synchro transmitters, and shall indicate or display the information data received. Each single (special) indicator-transmitter shall be provided with three concentric pointers. Pointer markings, where specified, shall be white for all black pointers; and shall be black for all white pointers. Pointer markings shall be black or white as specified hereinafter. When the type designation specifies that dial illumination shall be incorporated in the indicator-transmitter, the illumination power supply shall be taken from the power supply of the type 1F or 23TR4 synchro receiver. Unless otherwise specified herein, each single (special) indicator-transmitter shall be provided with three terminal strips of 6 terminals each.

3.7.2.1 Indicator-transmitter, rudder angle-order. - The rudder angle-order indicator-transmitter shall be provided with a 6-inch dial. Except as specified herein, the indicator-transmitter shall embody the characteristics of the rudder angle indicator (double), specified in 3.5.2.7, and the characteristics of the rudder order transmitter specified in 3.6.1.5. Pointer combination G or H shall be used as specified by the type. The terminal markings shall be as specified in 3.5.3.2.3, except that the signal push button shall be connected across terminals L and L4.

3.7.2.1.1 Size and weight. - The overall dimensions of the indicator-transmitter, including all knobs, and the weight of the indicator-transmitter shall not exceed the following:

Bulkhead, panel, and universal mountings

Height	12 inches
Width	10 inches
Depth	9-1/2 inches
Weight	45 pounds

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3.7.2.1.1.1 Pedestal mounting. - The overall dimensions of the indicator-transmitter provided with a pedestal mounting shall be in accordance with 3.7.2.1.1, and Specification MIL-I-983. The combined weight of the indicator-transmitter and the associated pedestal shall not exceed 110 pounds.

3.7.2.1.2 Pointers. - The outer and middle pointers of the indicator-transmitter shall be in accordance with the requirements for the outer and inner pointers, respectively, specified in 3.5.2.7.3.1. The inner pointer shall be in accordance with the pointer requirements for the transmitter specified in 3.6.1.5.

Note. - Cable for connection to ship's wiring - A type MSCA-19 cable (see Spec. MIL-C-2194) or a type MHFA-19 cable (see Spec. MIL-C-915) will be brought into the bottom of the instrument for connection of ship's wiring. The rudder angle-order indicator-transmitters now active are as follows:

Active types

IC/10SB48
IC/10UE48

3.7.2.2 Indicator -transmitter, engine order (handle operated). - The engine order indicator-transmitter shall be provided with 8-inch dials and shall incorporate in a single drum-shaped enclosure one transmitter and two indicators. The operating mechanism, pilot light, terminal markings and pointers shall conform to 3.7.1.1.5.1, 3.7.1.1.6, 3.7.1.2.3.1 and 3.7.3.2.3, respectively.

3.7.2.2.1 Size and weight. - The overall dimensions of the indicator-transmitter, including handle and pedestal, and the weight of the indicator-transmitter shall not exceed the following:

Pedestal mounting

Height	50 inches
Width	12-1/2 inches
Depth	12-1/2 inches
Weight	105 pounds

3.7.2.2.2 Enclosure. - The enclosure shall be of a general drum shape (exclusive of mounting base, etc.), with a transmitter operating handle of the lever type, located midway between the instrument faces (and external to the enclosure) and attached to the pointers required for indication of the transmitted order. A dial shall be incorporated in each end of the drum in accordance with figure 1 of Drawing 9000-S6504-73687 for the starboard dial and figure 2 for the port dial. The axis of rotation of the operating handles and the axis of each dial shall coincide with the axis of the drum.

3.7.2.2.3 Audible signals. - One type B1S4 or B1F4 bell shall be attached to the indicator-transmitter pedestal.

3.7.2.2.4 Handle locking device. - The handle of the indicator-transmitter shall be provided with a spring loaded locking device to lock the transmitter pointer in the desired order position. The locking device shall require the operation of a release by a positive action before the position of the handle can be changed.

Note. - Cable for connection to ship's wiring - A type MSCA-14 cable (see Spec. MIL-C-2194) or a type MHFA-14 cable (see Spec. MIL-C-915) will be brought into the bottom of the instrument for connection of ship's wiring. The engine order indicator-transmitters now active are as follows:

Active types

IC/10SC53

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3.7.3 Indicator-transmitters, double. - Each double indicator-transmitter shall receive two separate synchro signals from two separate remote synchro transmitters, and shall indicate or display the order or information data received. Each double indicator-transmitter shall also receive two separate mechanical inputs, and shall transmit two separate synchro signals corresponding to the mechanical inputs, and shall display the orders transmitted. Unless otherwise specified herein, each double indicator-transmitter shall be provided with two dials; and each dial shall be provided with two concentric pointers. One pointer of each dial shall indicate the received data; the other pointer of each dial shall indicate the transmitted data. The requirements specified in 3.7.1 for power supply for dial illumination shall apply. Unless otherwise specified herein, each double indicator-transmitter shall be provided with four terminal strips of 6 terminals each.

3.7.3.1 Indicator-transmitter, engine order (knob operated). - The engine order (knob operated) indicator-transmitter shall be provided with 4-inch dials or 6-inch dials, as designated by the type and shall incorporate, in a single enclosure, two indicator-transmitters identical to the indicator-transmitter with pointer combination 1 specified in 3.7.1.2, except as follows: (Two cables for audible signals will be furnished (see 3.5.2).)

3.7.3.1.1 Size and weight. - The overall dimensions of the indicator-transmitter, including all knobs, and the weight of the indicator-transmitter shall not exceed the following:

Bulkhead, panel, and universal mountings

Height	12 inches
Width	18 inches
Depth	9-1/4 inches
Weight	70 pounds

3.7.3.1.1.1 Pedestal mounting. - The overall dimensions of the indicator-transmitter provided with a pedestal mounting shall be in accordance with 3.7.3.1.1 and Specification MIL-I-983. The combined weight of the indicator-transmitter and the associated pedestal shall not exceed 135 pounds.

3.7.3.1.2 Terminal markings. - Terminals associated with the starboard (right hand) indicator-transmitter shall be marked in accordance with 3.7.1.2.3.1, except that each marking shall incorporate "1MB" in lieu of "MB". Terminals associated with the port (left hand) indicator-transmitter shall be marked in accordance with 3.7.1.2.3.1, except that each marking shall incorporate "2MB" in lieu of "MB".

Note. - Cable for connection to ship's wiring - A type MSCA-30 cable (see Spec. MIL-C-2194) or a type MHFA-30 cable (see Spec. MIL-C-915) will be brought into the bottom of the instrument for connection of ship's wiring. The engine order (knob operated) indicator-transmitters now active are as follows:

Active types

IC/8SB43
IC/8UB42

3.7.3.2 Indicator-transmitter, engine order (handle operated). - The engine order (handle operated) indicator-transmitter shall be provided with 8-inch dials and shall incorporate, in a single drum-shaped enclosure, two indicator-transmitters. The operating mechanism, pilot light, dial marking and terminal marking shall conform to 3.7.1.1.5.1, 3.7.1.1.6, 3.7.2.2.2 and 3.7.3.1.2, respectively. (Cables for connection to ships wiring will be as shown in note following 3.7.3.1.2.)

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3.7.3.2.1 Size and weight. - The overall dimensions of the indicator-transmitter, including handles and pedestal, and the weight of the indicator-transmitter shall not exceed the following:

Pedestal mounting

Height	49 inches
Width	12 inches
Depth	12 inches
Weight	105 pounds

3.7.3.2.2 Enclosure. - The enclosure shall be of a general drum-shape (exclusive of mounting base, etc.) with transmitter operating handles of the lever type mounted on the ends of the drum. A dial shall be incorporated in each end of the drum in accordance with figure 1 of Drawing 9000-S6504-73687 for the starboard dial and figure 2 for the port dial. The axis of rotation of the operating handles, and the axis of each dial, shall coincide with the axis of the drum.

3.7.3.2.3 Pointers. - A center pointer, of the revolving type, shall be incorporated for each of the two dials to indicate the received acknowledgement. The axis of rotation of each of the pointers shall coincide with the axis of the drum. The color of the pointers shall be in accordance with Specification MIL-I-983. The pointers required for indication of the transmitted orders shall be an integral part of the operating handles, and shall be located outside of the instrument windows along with the operating handles. The color of these pointers shall be the same as that of the operating handles.

3.7.3.2.4 Audible signals. - Two type B1S4 or B1F4 bells shall be attached to the indicator-transmitter pedestal.

3.7.3.2.5 Handle locking device. - The handles of the indicator-transmitter shall be provided with a spring-loaded locking device to lock the transmitter pointer in the desired order position. The locking device shall require the operation of a release by a positive action before the position of the handle can be changed.

Note. - The engine order (handle operated) indicator-transmitters now active are as follows:

Active types

IC/8SC53

3.7.3.3 Indicator-transmitter, motor order. - The motor order indicator-transmitter shall be identical to the indicator-transmitter specified in 3.7.3.1, except for dial markings and the additional detent positions required for the increase in active segments of the dial. The dial marking shall conform to 3.7.1.3.1.

Note. - The motor order indicator-transmitters now active are as follows:

Active types

IC/8UB42

3.7.4 Indicator-transmitters, triple. - Each triple indicator-transmitter shall receive three separate synchro signals from three separate remote synchro transmitters, and shall indicate or display the order or information data received. Each triple indicator-transmitter shall also receive three separate mechanical inputs, and shall transmit three separate synchro signals corresponding to the mechanical inputs, and shall display the orders transmitted. Unless otherwise specified herein, each triple indicator-transmitter shall be provided with six terminal strips of six terminals each.

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3.7.4.1 Indicator-transmitter, propeller order. - The propeller order indicator-transmitter shall be provided with six dials of the rotating disc type, designated as special dial size by the type. (One cable for audio signal will be furnished (see 3.5.1). The cables for connection to ships wiring will be as shown in note following 3.7.3.1.2.)

3.7.4.1.1 Size and weight. - The overall dimensions of the indicator-transmitter, including all knobs, and the weight of the indicator-transmitter shall not exceed the following:

Bulkhead, panel, and universal mountings

Height	10 inches
Width	14 inches
Depth	10 inches
Weight	55 pounds

3.7.4.1.1.1 Pedestal mounting. - The overall dimensions of the indicator-transmitter provided with a pedestal mounting shall be in accordance with 3.7.4.1.1 and Specification MIL-I-983. The combined weight of the indicator-transmitter and the associated pedestal shall not exceed 120 pounds.

3.7.4.1.2 Dials. - The indicator-transmitter shall be provided with three dials of the rotating disc type for the indicator portion of the instrument, namely, a separate dial for "hundreds", "tens", and "units"; and three corresponding dials for the transmitter portion of the instrument.

3.7.4.1.2.1 Dial numerals. - Numerals "0" to "9" inclusive, shall be arranged around each dial in a counterclockwise direction for increasing values. The dials for the indicator and transmitter shall be arranged in accordance with figures 11 and 12, respectively, of Drawing 9000-S6504-73687. The numerals shall not be less than 3/4 inch in height. Indicator and transmitter dials shall be arranged with respect to each other in accordance with the arrangement indicated in figures 11 and 12 of Drawing 9000-S6504-73687.

3.7.4.1.3 Instrument windows. - The indicator-transmitter shall be provided with three windows, one each for "hundreds", "tens", and "units". The left window shall display "hundreds" numerals, the center window "tens" numerals, and the right window "units" numerals. A masking plate shall be incorporated between each window and the associated disc type dials. The size of the openings in each masking plate shall be such as to display a single numeral on each of the associated dials, and exclude display of either adjacent numeral on each of the dials. The color of the masking plates shall be the same as the color of the dials. The instrument window openings shall be rectangular in shape, and shall be of such size as to show a 1/4 inch masking plate border around each opening in the masking plate. The arrangement of each instrument window shall be in general accordance with the window arrangement shown in conjunction with figures 11 and 12 of Drawing 9000-S6504-73687. The word "indicator" shall be incorporated in each masking plate immediately above the opening associated with the indicator dial; and the word "transmitter" shall be incorporated in each masking plate immediately below the opening associated with the transmitter dial.

3.7.4.1.4 Synchros. - Each indicator dial shall be positioned directly by a type 1F or type 23TR4 synchro receiver. Type 5G or 37TX4 synchro transmitters shall be used for transmitting the propeller order, and each synchro transmitter shall be positioned in conjunction with the positioning of the associated transmitter dial.

3.7.4.1.5 Operating knobs. - Each synchro transmitter and the associated transmitter dial shall be positioned by means of an operating knob. Clockwise rotation of each operating knob shall result in clockwise rotation of the associated transmitter dial. The ratio of angular movement and operating mechanism shall conform to 3.6.2.2.2.2 and 3.6.2.2.2.3, respectively.

3.7.4.1.5.1 Location. - Each operating knob shall be located directly below its respective instrument window opening; and shall be located so that not more than 20 percent of the height of the active numeral on the associated transmitter dial is ever obscured by the knob.

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3.7.4.1.6 Audible signals. - When a pedestal mounting is designated for the indicator-transmitter by the type specified, an audible signal shall be attached to the indicator-transmitter in accordance with the following:

3.7.4.1.6.1 Indicator-transmitter provided with dial illumination. - A type Z1S4 or Z1F4 buzzer shall be attached.

3.7.4.1.6.2 Indicator-transmitter not provided with dial illumination. - A type B3S4 or B3F4 bell shall be attached.

3.7.4.1.7 Internal assembly. - The internal assembly of the indicator-transmitter shall consist of 3 separate units. Each unit shall be capable of being removed from the instrument enclosure without disturbing the operation of the remaining units.

3.7.4.1.8 Terminal markings. -

3.7.4.1.8.1 Indicator-transmitter provided with dial illumination. - Terminals marked M-T, MM-T, M1, M2, and M3 shall be provided for connection to the synchro transmitter associated with the "hundreds" transmitter dial; and terminals marked M-I, MM-I, M11, M12, and M13 shall be provided for connection to the synchro receiver associated with the "hundreds" indicator dial. Similarly, terminals marked M-T, MM-T, M4, M5, and M6, and terminals marked M-I, MM-I, M14, M15, and M16 shall be provided for connection to the synchro transmitter and synchro receiver, respectively, associated with the "tens" transmitter and indicator dials, respectively. Similarly, terminals marked M-T, MM-T, M7, M8, and M9, and terminals M-I, MM-I, M17, M18, and M19 shall be provided for the synchro transmitter and receiver, respectively, associated with the "units" transmitter and indicator dials, respectively. Terminals marked M-T shall be connected together. Terminals marked MM-T shall be connected together. Terminals marked M-I shall be connected together. Terminals marked MM-I shall be connected together. Terminals marked M10 and M20 shall be provided for connection to the signal push button (and signal contacts specified in 3.7.4.1.5) and the audible signal, respectively. The signal push button incorporated in the indicator-transmitter shall be connected in parallel with the signal contacts specified in 3.7.4.1.5, and the signal devices shall then be connected across terminals M-T and M10. An audible signal when used in conjunction with the indicator-transmitter, shall be connected across terminals MM-I and M20.

3.7.4.1.8.2 Indicator-transmitter not provided with dial illumination. - Terminals marked M-T, MM-T, M11, M12, and M13 shall be provided for connection to the synchro transmitter associated with the "hundreds" transmitter dial; and terminals marked M-I, MM-I, M1, M2, and M3 shall be provided for connection to the synchro receiver associated with the "hundreds" indicator dial. Similarly, terminals marked M-T, MM-T, M14, M15, and M16, and terminals M-I, MM-I, M4, M5, and M6 shall be provided for connection to the synchro transmitter and synchro receiver, respectively, associated with the "tens" transmitter and indicator dials, respectively. Similarly, terminals marked M-T, MM-T, M17, and M18, and terminals marked M-I, MM-I, M7, M8, and M9 shall be provided for the synchro transmitter and receiver, respectively, associated with the "units" transmitter and indicator dials, respectively. Terminals marked M-T shall be connected together. Similar treatment shall be given to terminals marked MM-T, M-I, and MM-I in accordance with 3.7.4.1.8.1. Terminals marked M20 and M10 shall be provided for connection to the signal push button (and signal contacts specified in 3.7.4.1.5) and the audible signal, respectively. The signal push button incorporated in the indicator-transmitter shall be connected in parallel with the signal contacts specified in 3.7.4.1.5, and the signal devices shall then be connected across terminals M-T and M20. An audible signal, when used in conjunction with the indicator-transmitter shall be connected across terminals MM-I and M10.

Note. - The propeller order indicator-transmitters now active are as follows:

Active types
IC/9UF46

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3.8 Maintenance parts kits. - Unless otherwise specified in the contract or order, each maintenance parts kit, consisting of maintenance parts and special tools, shall be furnished on a per ship basis. Each maintenance parts kit shall contain the following items:

(a) One complete internal assembly of each type which is designed in accordance with 3.3.10.
(b) One instrument lighting transformer of each type.
(c) One dial window of each size and type.
(d) Twenty-five percent replacement of all special screws, nuts, studs, and similar items not carried in standard stock as defined in Specification MIL-I-983, and of parts frequently requiring renewal under operating conditions or subjected to considerable wear in operation.
(e) One set of special wrenches, gear pullers, or other special tools necessary for the care, maintenance, and adjustment of the equipment aboard ship. Requirements for the use of special tools shall be held to a minimum.
(f) Test cable assemblies, in accordance with Specification MIL-I-983 of types necessary to operate, test, and otherwise service each type of internal assembly associated with the equipment.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.2 Sampling procedures and tests. - The sampling procedures for the various tests to be performed on the equipment shall be as follows: (The attention of the manufacturer is invited to the applicable paragraphs of Specification MIL-I-983 for information concerning these tests, such as cost, where conducted, failures, test conditions, etc.)

4.2.1 Qualification. - Qualification tests shall be conducted at a laboratory designated by the Bureau of Ships. One each of the following ship control order and indicating instruments (or equivalent except for another permissible enclosure material) shall be subjected to qualification tests.

Indicator, rudder angle, type IC/1UA11
Indicator, engine order, type IC/2UA01
Transmitter, rudder angle, type IC/4BG20
Indicator-transmitter, engine order, type IC/7UE44
Indicator-transmitter, rudder angle-order, type IC/7SA48

These tests shall be performed, in general, in the order listed.

Test	Requirement reference	Test reference
General inspection	3.2.1	4.3.1
Supply line voltage and frequency	3.4.1	MIL-I-983
Radio frequency noise interference	3.2.1	MIL-I-983
Audible ambient noise	3.2.1	MIL-I-983
Oscillations	3.4.6	4.3.3
Temperature and humidity	3.2.1	MIL-I-983
Inclination	3.2.1	MIL-I-983
Dielectric strength	3.2.1	MIL-I-983
Insulation resistance	3.2.1	MIL-I-983
Accuracy	3.4.5	4.3.2
Enclosure	3.3.1	MIL-I-983
Salt spray	3.2.1	MIL-I-983
Vibration	3.2.1	MIL-I-983
Depth charge (applicable only to pressureproof instruments)	3.5.1.6.4	MIL-I-983
Shock	3.2.1	MIL-I-983

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4.2.2 Preliminary. - The first production unit of each type equipment covered in the contract or order shall be subjected to the following preliminary tests at a laboratory designated by the bureau or agency concerned. These tests will be required in the contract or order only where the successful bidders products have not passed the qualification tests specified in 4.2.1. These tests shall be performed, in general, in the order listed:

Test	Requirement reference	Test reference
General inspection	3.2.1	4.3.1
Supply line voltage and frequency	3.4.1	MIL-I-983
Radio frequency noise interference	3.2.1	MIL-I-983
Audible noise (airborne only)	3.2.1	MIL-I-983
Oscillations	3.4.6	4.3.3
Temperature and humidity	3.2.1	MIL-I-983
Inclination	3.2.1	MIL-I-983
Dielectric strength	3.2.1	MIL-I-983
Insulation resistance	3.2.1	MIL-I-983
Accuracy	3.4.5	4.3.2
Enclosure	3.3.1	MIL-I-983
Salt spray	3.2.1	MIL-I-983
Vibration	3.2.1	MIL-I-983
Depth charge (applicable only to pressureproof instruments)	3.5.1.6.4	MIL-I-983
Shock	3.2.1	MIL-I-983

4.2.3 Inspection tests at the place of manufacture. - A sample number of equipments shall be selected in accordance with 4.2.3.1.2, 4.2.3.1.3 and 4.2.3.1.4 and each of the selected equipments shall be subjected to the following tests at the place of manufacture, except that group C tests shall be conducted at a Government laboratory designated by the bureau or agency concerned. The tests shall be performed in general in the order listed and shall be conducted or witnessed by the Government inspector.

Test	Requirement reference	Test reference
Group A tests		
General inspection	3.2.1	4.3.1
Dielectric strength	3.2.1	MIL-I-983
Insulation resistance	3.2.1	MIL-I-983
Group B tests		
Supply line voltage and frequency variation	3.4.1	MIL-I-983
Oscillations	3.4.6	4.3.3
Group C tests		
Radio frequency noise interference	3.2.1	MIL-I-983
Audible noise (airborne only)	3.2.1	MIL-I-983
Temperature and humidity	3.2.1	MIL-I-983
Inclination	3.2.1	MIL-I-983
Accuracy	3.4.5	4.3.2

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Test	Requirement reference	Test reference
Group C tests (cont'd.)		
Enclosure	3.3.1	MIL-I-983
Salt spray	3.2.1	MIL-I-983
Vibration	3.2.1	MIL-I-983
Depth charge (applicable only to pressureproof instruments)	3.5.1.6.4	MIL-I-983
Shock	3.2.1	MIL-I-983

4.2.3.1 Sampling for inspection tests. -

4.2.3.1.1 Inspection lot. - All equipments of the same type presented for a Government acceptance inspection at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period at the choice of the Government inspector. Prior to presenting the lot for inspection, the contractor shall subject each equipment to the group A and B tests and shall include in the lot presented for Government inspection only those equipments that have passed these tests.

4.2.3.1.2 Sampling for group A tests. - A sample of equipments shall be selected from each inspection lot by the Government inspector in accordance with table VII and shall be subjected to each of the group A tests listed in 4.2.3 with lot acceptance based on the sampling requirements of Standard MIL-STD-105.

Table VII - Sampling for group A tests
AQL (approx.) = 1.5 percent defective.

Number of equipments in inspection lot	Number of equipments in sample	(Number of equipments nonconforming on any group A test)	
		Acceptance number	Rejection number
7 and under	All	-	-
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

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4.2.3.1.3 Sampling for group B tests. - A sample of equipments shall be selected from each inspection lot by the Government inspector in accordance with table VIII and shall be subjected to each of the group B tests listed in 4.2.3 with lot acceptance based on sampling requirements of Standard MIL-STD-105.

Table VIII - Sampling for group B tests.

Number of equipments in inspection lot	Number of equipments in sample	(Number of equipments nonconforming on any group B test)	
		Acceptance number	Rejection number
3 and under	All	-	-
4 to 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.2.3.1.4 Sampling for group C tests. - Group C tests will be required by the bureau or agency concerned only when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected by the Government inspector and subjected to each of the group C tests specified in 4.2.3 with acceptance based on 4.2.3.1.6.

4.2.3.1.5 Acceptance - group A and B tests. - The results of each test shall be compared with specification requirements. Failure to conform to this specification for any test shall be counted as a defect and the equipment shall be rejected. If the number of such nonconforming equipments in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection tests provided all equipments in the lot have been retested by the contractor for the test(s) causing rejection and all noncompliance corrected.

4.2.3.1.6 Acceptance - group C tests. - The results of each test shall be compared with specification requirements. In the event of failure to conform with this specification for any group C test, the contractor shall correct the cause of failure on future production units and repair the deficiency in all equipments produced on the contract or order.

4.3 Test procedures. -

4.3.1 General inspection. - The general inspection shall be conducted as specified in Specification MIL-I-983. The operating test shall consist of the connection of, and testing as a system, instruments assembled and connected in accordance with the loading characteristics of synchro units as specified in 3.4.2.1. The system shall be set up in accordance with the maximum loading limitations specified in 3.4.2.1. Instruments shall be used as required to make up such systems, and allowances shall be made for nonagreement of instrument dials among various types of instruments which may be assembled to comprise a test system. Instruments conforming a complete system for the operating test shall be connected with all associated auxiliary equipment, such as bells and buzzers, and operated at least 20 times without failure of a single item of equipment. Failure of any item of equipment shall require rectification of the item of the equipment by the contractor, and a repetition of the operating test.

4.3.2 Accuracy. - All indicator pointers, indicator rotating disc dials, etc., shall be tested for compliance with 3.4.5. Readings shall be observed at each 30 degree (geometrical) increment, clockwise and counterclockwise, around the complete dial.

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4.3.3 Oscillation. - When the electrical power supply is disconnected and the synchro transmitter rotor is off-set mechanically and locked at positions of 5, 10, 20, 90, and 180 geometrical degrees out of synchronism with the indicators, the elapsed time between the connection of the electrical supply and the time each indicator pointer comes to rest shall comply with 3.4.6.

5. PREPARATION FOR DELIVERY

5.1 Basic equipment and accessories. - The cleaning, preservation, packaging and packing shall be as specified hereinafter. Intended use (see 5.1.1 or 5.1.2) shall be specified (see 6.1).

5.1.1 Immediate use. - Equipment and accessories destined for shipment to domestic shipyards for immediate installation or use shall be cleaned, preserved, packaged and packed as follows:

5.1.1.1 Cleaning, preservation and packaging. - The cleaning, preservation, and packaging shall be in accordance with the manufacturers' commercial practice.

5.1.1.2 Packing. - Equipment and accessories shall be packed in accordance with the latest edition of the Consolidated Freight Classification Rules in effect on the date of invitation to bid. Shipping containers shall be of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to the point of delivery.

5.1.2 Stock, storage or overseas shipment. - Equipment and accessories destined for stock, storage, or overseas shipment shall be cleaned, preserved, packaged and packed as follows:

5.1.2.1 Cleaning, preservation, and packaging. - Equipment and accessories scheduled for stock, storage, or overseas shipment shall be cleaned by method C-0 and packaged by method II in accordance with Specification MIL-P-116.

5.1.2.2 Packing. -

5.1.2.2.1 For domestic shipment and storage. - Equipment and accessories destined for ultimate installation on a specific ship but temporarily stored shall be packed in wood cleated fiberboard, cleated plywood, nailed wood, corrugated or solid fiberboard boxes or nailed wood crates conforming to Specification NN-B-591, NN-B-601, NN-B-621, LLL-B-631, LLL-B-636 or MIL-C-6057, respectively. Closure of the fiberboard boxes shall be made with adhesive, metal fastenings or tape or a combination of these methods, in such a manner that the closures of the boxes shall not break or open when the box is tested in accordance with the applicable container specification. The gross weight of wood boxes shall not exceed 150 pounds unless the weight of the individual item exceeds this amount. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.1.2.2.2 Stock or overseas shipment. - Equipment and accessories destined for domestic shipment to stock or for overseas shipment shall be packed in wood-cleated-plywood, nailed wood or fiberboard boxes or nailed wood crates conforming to style A or B of Specification JAN-P-105, Specification JAN-P-106, or symbol V3c or V3s of Specification JAN-P-108 or Specification JAN-P-104, respectively. Plywood shall conform to type A or B, condition I of Specification JAN-P-139. Boxes shall be lined with a sealed waterproof case liner made of material conforming to Specification JAN-P-125. Liners shall be sealed with an adhesive conforming to Specification MIL-A-140. Seams and closure shall have strength and water-resistance equal to that of the body material or shall have a continuous seam of at least 3/4 inch wide. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. Flat steel strapping shall conform to class A or B of Specification QQ-S-781. The gross weight of wood boxes shall not exceed 150 pounds unless the weight of the unit item exceeds this amount. The gross weight of fiberboard boxes shall not exceed 70 pounds.

5.2 Repair parts. -

5.2.1 Cleaning, preservation and packaging. - Repair parts shall be cleaned, preserved and packaged in accordance with Specification MIL-P-116, using the applicable requirements of Specification JAN-P-658 as a guide.

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5.2.2 Packing. - Where onboard repair parts and tools are destined for a ship that is not fitted with bin or drawer type stowage, (see 6.1), the onboard repair parts and tools shall be packed in type M metal boxes conforming to Specification MIL-B-233 and shall be overpacked in shipping containers conforming to Specification NN-B-621 or in a nailed wood crate conforming to Specification JAN-P-132. Where onboard repair parts and tools are destined for a ship that is fitted with bin or drawer type stowage, (see 6.1), both onboard and stock repair parts shall be packed in wood cleated plywood, nailed wood, corrugated or solid fiberboard boxes conforming to Specification NN-B-601, NN-B-621, LLL-B-631, or LLL-B-636, respectively. Fiberboard boxes shall conform to the special requirements of the applicable box specification. Closure of the fiberboard boxes shall be made with adhesive, metal fastenings or tape or a combination of these methods, in such a manner that the closures of the boxes shall not break or open when the box is tested in accordance with the applicable container specification. The gross weight of wood boxes shall not exceed 150 pounds unless the weight of the individual item exceeds this amount. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.3 Publications. -

5.3.1 Packaging. - Instruction books and repair parts list shall be packaged in accordance with Specification MIL-B-15071.

5.3.2 Packing. - Two copies of the instruction book shall be packed within the shipping container holding the ship control order and indicating equipment. Bulk quantities of instruction books shall be packed in accordance with Specification MIL-B-15071. One copy of the repair parts list shall be packed in each type M box.

5.4 Marking. -

5.4.1 Interior containers. - In addition to the requirements specified in Standard MIL-STD-129, each interior package shall be clearly and legibly marked as follows:

(a) Electrical and operational characteristics (as applicable).

5.4.2 Exterior containers. - In addition to the marking specified in 5.4.1 and any special marking required by the contract or order, shipments shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type designation of equipment required (see 1.2).
(c) Material required for cast or sheet metal enclosure (see 1.2.3).
(d) Whether or not the preliminary tests are to be required (see 4.2.2).
(e) Intended use of equipment (i.e. immediate use, stock, storage or overseas shipment) as defined in 5.1.
(f) Type of stowage for onboard repair parts (see 5.2.2).

6.2 Qualification. - In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Military Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to communicate with the Bureau of Ships, Navy Department, Washington 25, D.C., and arrange to have the products that they propose to offer to the Army, the Navy, or the Air Force, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Navy Department, Washington 25, D.C.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-0270

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